

Elementary Number Theory

Handkerchief

Pratyay Sen

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§1 Well-Ordering and Mathematical Induction

We begin with two equivalent principles that underlie many proofs in number theory. It would not be an exaggeration to say that the entire branch of number theory would not exist without these two. Another interesting fact is that both of these are equivalent to one of my favourite axioms, the Axiom of Choice (more on that in the set theory section of this book). Let us first see what these two principles are, and how they are equivalent. Throughout this section, let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Well-ordering principle: Every nonempty subset $S \subseteq \mathbb{N}_0$ has a least element: there is an $m \in S$ such that $m \leq s$ for every $s \in S$.

Principle of mathematical induction: Let $P(n)$ be a statement about $n \in \mathbb{N}_0$. If

1. $P(0)$ holds (the *base case*), and
2. for every $k \in \mathbb{N}_0$, $P(k)$ implies $P(k+1)$ (the *inductive step*),

then $P(n)$ holds for every $n \in \mathbb{N}_0$.

Theorem:

The well-ordering principle and the principle of mathematical induction are equivalent.

§1.1 Well-Ordering Implies Induction

Suppose the well-ordering principle holds, and let $P(n)$ satisfy the base case and inductive step. Assume, for a contradiction, that $P(n)$ is false for at least one nonnegative integer. Then the set of counterexamples

$$C = \{n \in \mathbb{N}_0 : P(n) \text{ is false}\}$$

is nonempty, so it has a least element m . Since $P(0)$ is true, $m \neq 0$, and hence $m-1 \in \mathbb{N}_0$. Minimality of m implies that $P(m-1)$ is true. The inductive step now gives $P(m)$, contradicting $m \in C$. Thus there are no counterexamples, and $P(n)$ holds for every $n \in \mathbb{N}_0$. $\square$

§1.2 Induction Implies Well-Ordering

Suppose the principle of mathematical induction holds. Let $S \subseteq \mathbb{N}_0$ be nonempty, and assume, for a contradiction, that S has no least element. Consider the statement

$$Q(n) : \quad S \cap \{0, 1, \dots, n\} = \emptyset.$$

We prove $Q(n)$ by induction.

Base case: If $0 \in S$, then 0 would be its least element. Hence $0 \notin S$, so $Q(0)$ holds.

Inductive step: Assume $Q(k)$ holds. If $k+1 \in S$, then, since none of $0, 1, \dots, k$ belongs to S , the integer $k+1$ would be the least element of S . This contradicts our assumption. Therefore $k+1 \notin S$, and $Q(k+1)$ holds.

By induction, $Q(n)$ holds for every $n \in \mathbb{N}_0$. But choosing any $s \in S$ contradicts $Q(s)$. Thus S must have a least element, which proves the well-ordering principle. $\square$

§1.3 Strong Induction

An equivalent version of induction, is *strong induction*: if $P(0)$ holds and, for every $k \geq 0$,

$$P(0), P(1), \dots, P(k) \implies P(k+1),$$

then $P(n)$ holds for every $n \in \mathbb{N}_0$.

Ordinary induction proves this by applying it to the cumulative statement $R(n)$ that $P(j)$ holds for every $0 \leq j \leq n$. Conversely, an ordinary inductive step is a special case of a strong inductive step, since the latter includes $P(k)$ among its assumptions. Thus well-ordering, ordinary induction, and strong induction are interchangeable proof principles. All of these are very useful in number theory for proving a variety of results.

§2 Divisibility

Divisibility sits at the very core of number theory, because the very notion of a number dividing another gave birth to modular arithmetic, the concept of primes, and basically the entirety of number theory.

For integers a and b , we write $a \mid b$ if there exists an integer k such that $b = ak$.

For example, $7 \mid 42$, but $7 \nmid 45$.

If $a \mid b$ and $a \mid c$, then

$$a \mid (ub + vc) \quad \text{for every } u, v \in \mathbb{Z}.$$

Proof: Since $a \mid b$ and $a \mid c$, there exist integers r, s such that $b = ar$ and $c = as$. For any integers u, v , we therefore have

$$ub + vc = u(ar) + v(as) = a(ur + vs).$$

Because $ur + vs \in \mathbb{Z}$, the definition of divisibility gives $a \mid (ub + vc)$. □

The division algorithm states that for every integer a and positive integer b , there are unique integers q, r with

$$a = bq + r, \quad 0 \leq r < b.$$

Proof of existence: Consider the set of nonnegative candidate remainders

$$S = \{a - bk : k \in \mathbb{Z}, a - bk \geq 0\}.$$

This set is nonempty: taking $k = -|a|$ gives $a + b|a| \geq a + |a| \geq 0$, since $b \geq 1$. By the well-ordering principle, S has a least element r . By definition of S , there is an integer q with $r = a - bq$, so $a = bq + r$ and $r \geq 0$.

If $r \geq b$, then

$$r - b = a - b(q + 1) \geq 0,$$

so $r - b$ would also belong to S . But $b > 0$ implies $r - b < r$, contradicting the minimality of r . Therefore $0 \leq r < b$.

Proof of uniqueness: Suppose there are two representations

$$a = bq + r = bq' + r', \quad 0 \leq r, r' < b.$$

Subtracting yields

$$b(q - q') = r' - r.$$

Since $-b < r' - r < b$, the integer $r' - r$ is a multiple of b with absolute value strictly less than b . The only such multiple is 0. Thus $r' = r$, and then $b(q - q') = 0$ gives $q = q'$ because $b > 0$. This proves both existence and uniqueness, including when a is negative. $\square$

We'll be using well ordering, induction and divisibility throughout this section, so make sure you remember them!! (or write them in a piece of paper)

§3 GCD and LCM

After looking at divisibility, a natural question one might ask is what about the common divisors of two numbers? Or the common multiples? It is obvious that the lowest common divisor for any two numbers is 1, and the greatest common multiple is infinity. Therefore, it makes sense to talk about the greatest common divisor and least common multiple.

§3.1 Greatest Common Divisors

For integers a, b that are not both zero, their greatest common divisor, written $\gcd(a, b)$, is the largest positive integer dividing both.

If $\gcd(a, b) = 1$, we say that a and b are *coprime* or *relatively prime*.

Euclid had developed a very useful algorithm in order to find the gcd of any two numbers, because prime factorisation is not a suitable method all the time.

Euclidean Algorithm

For integers a, b not both zero, repeated division computes $\gcd(a, b)$: the algorithm terminates, and its last nonzero remainder is the greatest common divisor. If $a = bq + r$ with $b > 0$ and $0 \leq r < b$, then

$$\gcd(a, b) = \gcd(b, r).$$

Indeed, any common divisor of a, b divides $r = a - bq$. Conversely, any common divisor of b, r divides $a = bq + r$. Thus the two pairs have exactly the same positive common divisors, and hence the same gcd.

Procedure

Replace the inputs by their absolute values and, if necessary, interchange them so that $r_0 \geq r_1 \geq 0$, with $r_0 > 0$. These operations do not change the gcd. If $r_1 = 0$, return r_0 , since $\gcd(r_0, 0) = r_0$. Otherwise, repeatedly apply the division algorithm:

$$r_{j-1} = q_j r_j + r_{j+1}, \quad 0 \leq r_{j+1} < r_j,$$

for $j = 1, 2, \dots$, stopping as soon as $r_{j+1} = 0$.

Proof of termination: If the procedure never stopped, it would produce an infinite strictly decreasing sequence of positive integers

$$r_1 > r_2 > r_3 > \cdots.$$

By well-ordering, the set of these remainders would have a least element r_k . But the next remainder r_{k+1} would be positive and smaller than r_k , a contradiction. Therefore, for some $n \geq 1$, we reach $r_n > 0$ and $r_{n+1} = 0$.

Proof of correctness: The basic step preserves the gcd at every division. Hence

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n.$$

Thus the last nonzero remainder is exactly the gcd of the original inputs. $\square$

Example: To find $\gcd(252, 105)$, perform the divisions

$$252 = 2 \cdot 105 + 42,$$

$$105 = 2 \cdot 42 + 21,$$

$$42 = 2 \cdot 21.$$

The last nonzero remainder is 21, so $\gcd(252, 105) = 21$.

Bézout's theorem:

For integers a, b not both zero, there exist integers u, v such that

$$au + bv = \gcd(a, b).$$

Proof: Consider the set of positive integer linear combinations of a and b ,

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}.$$

Since a, b are not both zero, at least one of $|a|$ and $|b|$ is positive and belongs to S . Thus S is nonempty. By well-ordering, S has a least element d , and we may write

$$d = au + bv > 0$$

for some integers u, v .

Apply the division algorithm to a with divisor d :

$$a = dq + r, \quad 0 \leq r < d.$$

Then

$$r = a - qd = a(1 - qu) + b(-qv).$$

If $r > 0$, this would be an element of S smaller than d , a contradiction. Hence $r = 0$, so $d \mid a$. Likewise, writing $b = dq' + r'$ with $0 \leq r' < d$ gives

$$r' = b - q'd = a(-q'u) + b(1 - q'v).$$

Minimality again forces $r' = 0$, so $d \mid b$.

Finally, every positive common divisor c of a and b divides $au + bv = d$. Since $c, d > 0$, this implies $c \leq d$. Therefore d is the greatest common divisor of a and b , proving $au + bv = \gcd(a, b)$. $\square$

These coefficients can be found by backward substitution in the Euclidean algorithm. In the example above,

$$21 = 105 - 2(252 - 2 \cdot 105) = -2 \cdot 252 + 5 \cdot 105.$$

Thus begins the barrage of lemmas...

Lemma: For integers a, b not both zero,

$$\gcd(a, b) = 1 \iff \text{there exist } x, y \in \mathbb{Z} \text{ such that } ax + by = 1.$$

Here (a, b) is another common notation for $\gcd(a, b)$. The coefficients x, y are required to exist; the equation need not hold for arbitrary choices of x, y .

Proof: If $\gcd(a, b) = 1$, Bézout's identity immediately supplies integers x, y with $ax + by = 1$. Conversely, suppose $ax + by = 1$ for some integers x, y . Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, we have $d \mid (ax + by) = 1$. As d is positive, $d = 1$. This proves the equivalence. $\square$

For example, $3 \cdot 5 - 2 \cdot 7 = 1$, so $\gcd(5, 7) = 1$.

Lemma: Let a, b be integers, not both zero, and let $d = \gcd(a, b)$. Then

$$\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1.$$

Proof: Since $d > 0$ divides both a and b , the quotients a/d and b/d are integers. By Bézout's identity, there exist integers x, y such that

$$ax + by = d.$$

Dividing by d gives

$$\frac{a}{d}x + \frac{b}{d}y = 1.$$

The coprimality criterion therefore implies $\gcd(a/d, b/d) = 1$. $\square$

Lemma: Let a, b, c be integers with $\gcd(a, b) = 1$. If $a \mid c$ and $b \mid c$, then $ab \mid c$.

Proof: Since $a \mid c$ and $b \mid c$, there exist integers r, s such that $c = ar = bs$. By Bézout's identity, there exist integers x, y with $ax + by = 1$. Multiplying this identity by c gives

$$c = axc + byc = ax(bs) + by(ar) = ab(xs + yr).$$

Since $xs + yr$ is an integer, $ab \mid c$. $\square$

Why the coprimality assumption matters: Without $\gcd(a, b) = 1$, the conclusion need not hold. For example, $4 \mid 12$ and $6 \mid 12$, but $4 \cdot 6 = 24 \nmid 12$.

Euclid's lemma: Let a, b, c, d be integers with $\gcd(a, b) = 1$. If $ac = bd$, then $a \mid d$.

Proof: By Bézout's identity, there exist integers x, y such that $ax + by = 1$. Multiplying by d and using $bd = ac$, we obtain

$$d = axd + byd = a(xd) + y(bd) = a(xd) + y(ac) = a(xd + cy).$$

Since $xd + cy$ is an integer, $a \mid d$. □

Equivalently, if $\gcd(a, b) = 1$ and $a \mid bd$, then $a \mid d$: the divisibility $a \mid bd$ means precisely that $ac = bd$ for some integer c .

§3.2 Least Common Multiples

For positive integers a, b , the *least common multiple*, denoted $\text{lcm}(a, b)$ or $[a, b]$, is the smallest positive integer that is divisible by both a and b .

It exists because the set of positive common multiples is nonempty (it contains ab) and therefore has a least element by well-ordering.

For example, $[4, 6] = 12$ and $[8, 9] = 72$.

The gcd-lcm identity:

For positive integers a, b ,

$$ab = \gcd(a, b) \text{lcm}(a, b) = (a, b)[a, b].$$

Proof: Let $d = \gcd(a, b)$ and write $a = du$, $b = dv$, where u, v are positive integers. By the result on dividing out the gcd, $\gcd(u, v) = 1$. The positive integer

$$L = duv = av = bu$$

is a common multiple of a and b .

Now let m be any positive common multiple of a and b . Since $a \mid m$, we can write $m = ak = duk$ for a positive integer k . Also $b \mid m$, so $m = b\ell = dv\ell$ for a positive integer ℓ . Canceling d gives $uk = v\ell$, and hence $v \mid uk$. Since $\gcd(u, v) = 1$, Euclid's lemma implies $v \mid k$. Writing $k = vt$ with t a positive integer, we obtain

$$m = duvt = Lt.$$

Thus L divides every positive common multiple, so each is at least L . Since L is itself a positive common multiple, $[a, b] = L = duv$. Consequently,

$$(a, b)[a, b] = d \cdot duv = (du)(dv) = ab.$$

□

For example, $(12, 18) = 6$ and $[12, 18] = 36$, giving $12 \cdot 18 = 6 \cdot 36 = 216$.

For nonzero integers of either sign, define $[a, b] = [|a|, |b|]$. The general identity is then

$$(a, b)[a, b] = |ab|.$$

If exactly one input is zero, the convention $[a, 0] = [0, a] = 0$ extends this identity, since $(a, 0) = |a|$ for $a \neq 0$.

§4 Primes and Unique Factorization

We know that a number must have itself and 1 as divisors, so the natural question to ask here is what if a number has *only* 1 and itself as its divisors?

An integer $p > 1$ is *prime* if its only positive divisors are 1 and p . An integer greater than 1 that is not prime is *composite*. The number 1 is neither prime nor composite.

Euclid's lemma: If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: If $p \nmid a$, then $\gcd(p, a) = 1$. Bézout's identity gives integers u, v satisfying $up + va = 1$. Multiplying by b yields $upb + vab = b$. Both terms on the left are divisible by p , so $p \mid b$. $\square$

Fundamental theorem of arithmetic:

Every integer $n > 1$ can be written as a product of primes, uniquely up to the order of the factors.

For example,

$$360 = 2^3 \cdot 3^2 \cdot 5.$$

Proof of existence: We use strong induction on $n \geq 2$. The base case $n = 2$ holds because 2 is prime and is therefore already a product consisting of one prime.

Now let $n > 2$ and assume every integer m with $2 \leq m < n$ can be written as a product of primes. If n is prime, there is nothing to prove. If n is composite, it has a positive divisor a with $1 < a < n$. Writing $n = ab$, we also have $1 < b < n$. By the inductive hypothesis, both factors have prime factorizations, say

$$a = p_1 \cdots p_r, \quad b = q_1 \cdots q_s.$$

Their product gives a prime factorization of n :

$$n = p_1 \cdots p_r q_1 \cdots q_s.$$

Thus every integer $n > 1$ is a product of primes.

Proof of uniqueness: Suppose n has two prime factorizations

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where prime factors may repeat. Interchanging the two factorizations if necessary, assume $r \leq s$. Since $p_1 \mid q_1 q_2 \cdots q_s$, repeated application of Euclid's lemma shows that $p_1 \mid q_j$ for some j . Indeed, applying the lemma to the first factor and the product of the remaining factors, and continuing, must yield a factor divisible by p_1 . As q_j is prime and $p_1 > 1$, it follows that $p_1 = q_j$. Reorder the q -factors so that q_j comes first, and cancel this common nonzero factor to obtain

$$p_2 \cdots p_r = q_2 \cdots q_s.$$

Repeat the same argument for each remaining prime on the left. After r cancellations, if $r < s$ we would have

$$1 = q_{r+1} \cdots q_s,$$

which is impossible because the product on the right contains at least one prime and is therefore at least 2. Hence $r = s$, and after reordering, $p_j = q_j$ for every $j = 1, \dots, r$. The prime factorization is therefore unique up to order. $\square$

Grouping equal prime factors gives the equivalent form

$$n = p_1^{e_1} \cdots p_k^{e_k}, \quad p_1 < \cdots < p_k, \quad e_1, \dots, e_k \geq 1,$$

where the p_j are distinct primes and the exponents are integers. Both the ordered list of primes and their exponents are uniquely determined by n .

§4.1 Infinitely Many Primes

It seems natural that there would be infinitely many primes, but its not exactly trivial to prove, so here are some beautiful proofs of the theorem.

Theorem:

There are infinitely many prime numbers. We give three proofs, using respectively a new prime divisor, a sequence of pairwise coprime integers, and the divergence of the harmonic series.

Proof 1: Euclid's construction: Suppose there were only finitely many primes, listed as $p_1, \dots, p_k$. Consider

$$N = p_1 p_2 \cdots p_k + 1.$$

Since $N > 1$, it has a prime divisor q . If q were one of the listed primes, it would divide both N and $p_1 \cdots p_k$, and therefore their difference 1. This is impossible. Thus q is a prime not in the list, a contradiction. $\square$

The number N need not itself be prime; the argument only requires that it have a prime divisor. More generally, this construction finds a prime outside any given finite list of primes.

Proof 2: Pairwise coprime Fermat numbers:

For $n \geq 0$, define the *Fermat numbers* by

$$F_n = 2^{2^n} + 1.$$

We first prove the identity

$$F_0 F_1 \cdots F_{n-1} = F_n - 2 \quad (n \geq 1).$$

For $n = 1$, it reads $3 = 5 - 2$. If it holds for n , then

$$F_0 \cdots F_n = (F_n - 2)F_n = (2^{2^n} - 1)(2^{2^n} + 1) = 2^{2^{n+1}} - 1 = F_{n+1} - 2.$$

Thus the identity follows by induction.

If $0 \leq m < n$, then $F_m \mid F_n - 2$. Every positive common divisor of F_m and F_n therefore divides 2. But both Fermat numbers are odd, so their gcd is 1. Hence the Fermat numbers are pairwise coprime.

Each $F_n > 1$ has a prime divisor. No prime can divide two different Fermat numbers, because they are coprime. Consequently, these infinitely many integers supply infinitely many distinct primes. $\square$

Proof 3: A harmonic-series argument: Suppose all primes formed a finite list $p_1, \dots, p_k$, and put

$$C = \prod_{j=1}^k \frac{1}{1 - 1/p_j}.$$

This is a finite positive number. We will show that the harmonic sums

$$H_N = \sum_{n=1}^N \frac{1}{n}$$

would all be bounded by C .

By unique factorization, every $1 \leq n \leq N$ has a unique expression $n = p_1^{e_1} \cdots p_k^{e_k}$ with nonnegative integer exponents; $n = 1$ corresponds to all exponents being zero. Each $e_j \leq N$: indeed, $2^{e_j} \leq p_j^{e_j} \leq n \leq N$, and $e_j \leq 2^{e_j}$ for every nonnegative integer e_j . Expanding the following finite product therefore includes every term $1/n$ for $1 \leq n \leq N$, as well as possibly other positive terms:

$$H_N \leq \prod_{j=1}^k \left(\sum_{e=0}^N \frac{1}{p_j^e} \right) \leq \prod_{j=1}^k \frac{1}{1 - 1/p_j} = C.$$

The second inequality follows from the finite geometric-sum formula.

On the other hand, grouping the harmonic sum into blocks gives, for any positive integer m ,

$$H_{2^m} = 1 + \sum_{j=1}^m \sum_{n=2^{j-1}+1}^{2^j} \frac{1}{n} \geq 1 + \sum_{j=1}^m \frac{2^{j-1}}{2^j} = 1 + \frac{m}{2}.$$

Each block has 2^{j-1} terms, each at least $1/2^j$. The lower bound becomes arbitrarily large as m increases, contradicting $H_N \leq C$. Therefore there must be infinitely many primes. $\square$

§5 Modular Arithmetic

At long last, we jump into the system of congruences that makes our lives easier by us being able to write a number, its remainder and divisor in a single line, in a convenient manner. This new language of mathematics was developed by Gauss when he was just 19 years old.

§5.1 Congruences and Their Basic Properties

Fix a positive integer m . For integers a, b , define

$$a \equiv b \pmod{m} \iff m \mid (a - b).$$

By the division algorithm, this means that a and b have the same remainder upon division by m .

For example, $29 \equiv 5 \pmod{12}$ and $-1 \equiv 6 \pmod{7}$.

Modular arithmetic:

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \quad a - c \equiv b - d \pmod{m}, \quad ac \equiv bd \pmod{m}.$$

Proof: The modulus m divides $a - b$ and $c - d$, and therefore divides

$$(a + c) - (b + d) = (a - b) + (c - d), \quad (a - c) - (b - d) = (a - b) - (c - d).$$

It also divides $ac - bd = c(a - b) + b(c - d)$. $\square$

Repeated multiplication gives $a^r \equiv b^r \pmod{m}$ for every positive integer r .

Combining addition and multiplication shows that $P(a) \equiv P(b) \pmod{m}$ for every polynomial P with integer coefficients.

Also, if a positive integer d divides m , then $a \equiv b \pmod{m}$ implies $a \equiv b \pmod{d}$.

For example, $7^2 \equiv 1 \pmod{12}$ implies

$$7^{100} = (7^2)^{50} \equiv 1 \pmod{12}.$$

Cancellation:

For any integer c , put $g = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/g}.$$

Proof: Write $c = gc'$ and $m = gm'$. Since $\gcd(c', m') = 1$, Euclid's lemma gives

$$m \mid c(a - b) \iff m' \mid c'(a - b) \iff m' \mid (a - b).$$

□

In particular, a factor coprime to m can be canceled without changing the modulus. Without coprimality, this need not be valid: $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$, but $1 \not\equiv 4 \pmod{6}$. The correct cancellation in this example gives $1 \equiv 4 \pmod{3}$.

§5.2 Congruence as an Equivalence Relation

Now we shall slowly step in the waters of group theory...

Proposition:

Congruence modulo m is an equivalence relation on $\mathbb{Z}$.

Proof: We verify the three defining properties.

1. *Reflexivity:* $m \mid 0 = a - a$, so $a \equiv a \pmod{m}$.
2. *Symmetry:* if $m \mid (a - b)$, then $m \mid (b - a)$; hence $a \equiv b \pmod{m}$ implies $b \equiv a \pmod{m}$.
3. *Transitivity:* if $m \mid (a - b)$ and $m \mid (b - c)$, then $m \mid (a - c)$; thus $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ imply $a \equiv c \pmod{m}$.

□

The equivalence class of a is

$$[a]_m = \{b \in \mathbb{Z} : b \equiv a \pmod{m}\} = a + m\mathbb{Z}.$$

Two classes are equal exactly when their representatives are congruent. Indeed, $a \equiv b \pmod{m}$ makes membership in either class equivalent by transitivity; conversely, equality of the classes

implies $a \in [b]_m$. If two classes intersect, the same argument shows that they are equal. Since every integer belongs to its own class, the classes partition $\mathbb{Z}$.

The set of these classes is the quotient

$$\mathbb{Z}/m\mathbb{Z} = \{[0]_m, [1]_m, \dots, [m-1]_m\}.$$

Every integer has exactly one remainder in $\{0, \dots, m-1\}$, so there are exactly m classes.

For example, $[2]_5 = \{\dots, -8, -3, 2, 7, 12, \dots\}$ and $\mathbb{Z}/5\mathbb{Z} = \{[0]_5, [1]_5, \dots, [4]_5\}$.

The elements of $\mathbb{Z}/m\mathbb{Z}$ are classes, not individual integers, although their standard representatives are often used in calculations.

§5.3 Complete Residue Systems

Definition: A *complete residue system modulo m* is a set of integers containing exactly one representative of each class in $\mathbb{Z}/m\mathbb{Z}$. Equivalently, it consists of m integers that are pairwise incongruent modulo m . Thus every integer is congruent modulo m to exactly one member of the system.

The standard complete residue system is

$$\{0, 1, \dots, m-1\}.$$

Any m consecutive integers also form a complete residue system: the difference between two distinct members has absolute value less than m and cannot be divisible by m .

For example, modulo 5,

$$\{0, 1, 2, 3, 4\}, \quad \{1, 2, 3, 4, 5\}, \quad \{-2, -1, 0, 1, 2\}$$

are all complete residue systems. The representatives need not be positive or lie between 0 and $m-1$.

Multiplication preserves complete residue systems:

Let $r_1, \dots, r_m$ be a complete residue system modulo m , and let a be an integer with $\gcd(a, m) = 1$. Then

$$ar_1, \dots, ar_m$$

is also a complete residue system modulo m .

Proof: If $ar_i \equiv ar_j \pmod{m}$, cancellation of the coprime factor a gives $r_i \equiv r_j \pmod{m}$. Since the original system has no two congruent representatives, $i = j$. Thus the m products are pairwise incongruent modulo m . There are exactly m residue classes, so the products represent every class exactly once. $\square$

§5.4 Arithmetic on Residue Classes

Define

$$[a]_m + [b]_m = [a + b]_m, \quad [a]_m [b]_m = [ab]_m.$$

These operations are *well-defined*: replacing a, b by congruent representatives does not change the resulting classes, by compatibility with addition and multiplication. Associativity, commutativity, and distributivity follow from the corresponding integer identities. The additive identity is $[0]_m$, the additive inverse of $[a]_m$ is $[-a]_m$, and the multiplicative identity is $[1]_m$. Thus $\mathbb{Z}/m\mathbb{Z}$ is a commutative ring.

§5.5 Modular Inverses

Let $m \geq 2$. A multiplicative inverse of $[a]_m$ is a class $[u]_m$ such that $[a]_m [u]_m = [1]_m$, or equivalently $au \equiv 1 \pmod{m}$.

Theorem:

The class $[a]_m$ has a multiplicative inverse if and only if $\gcd(a, m) = 1$.

Proof: If $\gcd(a, m) = 1$, Bézout's identity supplies integers u, v with $au + mv = 1$. Reducing modulo m gives $au \equiv 1 \pmod{m}$. Conversely, if $au \equiv 1 \pmod{m}$, then $au - mv = 1$ for some integer v . Every common divisor of a, m divides the left side, hence divides 1. Therefore $\gcd(a, m) = 1$. $\square$ The inverse class is unique. If $au \equiv av \equiv 1 \pmod{m}$, then $u \equiv v \pmod{m}$ by cancellation, since a is coprime to m . There may be many integer representatives of this one inverse class.

For example, $7 \cdot 7 \equiv 1 \pmod{12}$, so

$$7x \equiv 5 \pmod{12} \iff x \equiv 7 \cdot 5 \equiv 11 \pmod{12}.$$

In contrast, $[6]_{12}$ has no inverse because $\gcd(6, 12) = 6$.

§6 Euler's Totient Function

§6.1 The Unit Group and Its Cardinality

For $m \geq 2$, the *unit group* is

$$(\mathbb{Z}/m\mathbb{Z})^* = \{[a]_m : [a]_m \text{ has a multiplicative inverse}\} = \{[a]_m : \gcd(a, m) = 1\}.$$

It is also denoted $(\mathbb{Z}/m\mathbb{Z})^\times$.

Multiplication makes this set an abelian group. Products of units are units, since an inverse of $[a]_m [b]_m$ is $[b]_m^{-1} [a]_m^{-1}$. The identity $[1]_m$ is a unit; inverses of units are units; and associativity and commutativity are inherited from the residue ring.

Definition: Euler's totient function is the cardinality of this group:

$$\varphi(m) = |(\mathbb{Z}/m\mathbb{Z})^*| = \#\{a : 1 \leq a \leq m, \gcd(a, m) = 1\}.$$

We set $\varphi(1) = 1$ by convention.

For example,

$$(\mathbb{Z}/12\mathbb{Z})^* = \{[1]_{12}, [5]_{12}, [7]_{12}, [11]_{12}\}, \quad \varphi(12) = 4.$$

§6.2 Reduced Residue Systems

Definition: For $m \geq 2$, a *reduced residue system modulo m* is a set of integers containing exactly one representative of each invertible class in $(\mathbb{Z}/m\mathbb{Z})^*$. Equivalently, it consists of $\varphi(m)$ integers, each coprime to m , that are pairwise incongruent modulo m . Every integer coprime to m is congruent to exactly one member of the system.

The standard reduced residue system is

$$\{a : 1 \leq a < m, \gcd(a, m) = 1\}.$$

For example, modulo 12, both

$$\{1, 5, 7, 11\} \quad \text{and} \quad \{13, 17, -5, -1\}$$

are reduced residue systems: their members represent the same four unit classes. By contrast, $\{1, 2, \dots, 11\}$ is not reduced modulo 12, because some of its members are not coprime to 12.

Deleting all members not coprime to m from any complete residue system produces a reduced residue system. This works because coprimality with m depends only on the residue class. For a prime p , the set $\{1, 2, \dots, p-1\}$ is therefore a reduced residue system modulo p . A complete system represents *all* m classes; a reduced system represents only the $\varphi(m)$ *unit* classes.

Multiplication preserves reduced residue systems:

Similar to complete residue systems, let $r_1, \dots, r_{\varphi(m)}$ be a reduced residue system modulo $m \geq 2$, and suppose $\gcd(a, m) = 1$. Then

$$ar_1, \dots, ar_{\varphi(m)}$$

is also a reduced residue system modulo m .

Proof: Every product ar_i is coprime to m : both $[a]_m$ and $[r_i]_m$ are units, so their product $[ar_i]_m$ is a unit as well. Equivalently, if $au \equiv 1 \pmod{m}$ and $r_i v_i \equiv 1 \pmod{m}$, then $(ar_i)(uv_i) \equiv 1 \pmod{m}$.

If $ar_i \equiv ar_j \pmod{m}$, cancellation gives $r_i \equiv r_j \pmod{m}$, and hence $i = j$. Thus the products give $\varphi(m)$ distinct unit classes. Since there are exactly $\varphi(m)$ such classes, they form a reduced residue system. $\square$

In both results, multiplication by a number coprime to m permutes the represented residue classes; it need not preserve the chosen integer representatives themselves.

§6.3 Computing the Totient

Formula:

For every integer $m \geq 2$,

$$\varphi(m) = m \prod_{p|m} \left(1 - \frac{1}{p}\right),$$

where the product is over the distinct prime divisors of m .

Proof: An integer is coprime to m exactly when none of these primes divides it. Count the integers in $\{1, \dots, m\}$ by inclusion–exclusion. For any collection of distinct prime divisors $p_1, \dots, p_j$ of m , exactly $m/(p_1 \cdots p_j)$ integers in this range are divisible by all of them. Subtracting the single-prime counts, adding the pair counts, and continuing gives precisely the expanded product above. $\square$

If p is prime, all $p - 1$ nonzero classes modulo p are units, so $\varphi(p) = p - 1$. More generally, for $r \geq 1$,

$$\varphi(p^r) = p^r - p^{r-1} = p^r \left(1 - \frac{1}{p}\right),$$

since exactly p^{r-1} integers among $1, \dots, p^r$ are divisible by p .

§6.4 Multiplicativity and a Divisor-Sum Identity

Multiplicativity of the totient:

If m, n are positive integers with $\gcd(m, n) = 1$, then

$$\varphi(mn) = \varphi(m)\varphi(n).$$

Proof: If either integer is 1, the result follows from $\varphi(1) = 1$. Otherwise, coprimality means that no prime divides both m and n . By Euclid's lemma, the primes dividing mn are precisely those dividing m or n . Thus the product formula splits into two disjoint products:

$$\begin{aligned} \varphi(mn) &= mn \prod_{p|mn} \left(1 - \frac{1}{p}\right) \\ &= \left(m \prod_{p|m} \left(1 - \frac{1}{p}\right)\right) \left(n \prod_{p|n} \left(1 - \frac{1}{p}\right)\right) \\ &= \varphi(m)\varphi(n). \end{aligned}$$

$\square$

The coprimality hypothesis is essential: taking $m = n = 2$ gives $\varphi(4) = 2$, whereas $\varphi(2)\varphi(2) = 1$.

A second proof using residue classes will follow from the Chinese remainder theorem.

Divisor-sum identity:

For every positive integer n ,

$$\sum_{d|n} \varphi(d) = n,$$

where the sum is over the positive divisors of n .

Proof: Partition the integers $1, 2, \dots, n$ according to their gcd with n . For each positive divisor d of n , put

$$A_d = \{a : 1 \leq a \leq n, \gcd(a, n) = n/d\}.$$

Every integer in the range belongs to exactly one such set, since its gcd with n is a unique positive divisor of n .

Write $g = n/d$. The map $b \mapsto gb$ is a bijection from

$$\{b : 1 \leq b \leq d, \gcd(b, d) = 1\} \quad \text{onto} \quad A_d.$$

Indeed, if $\gcd(a, n) = g$, dividing out the gcd gives $a = gb$, $n = gd$, with $1 \leq b \leq d$ and $\gcd(b, d) = 1$. Conversely, if $\gcd(b, d) = 1$, choose integers u, v with $bu + dv = 1$. Then $g = (gb)u + (gd)v$, so every common divisor of gb, gd divides g . Since g itself divides both, $\gcd(gb, gd) = g$. The map is injective because $g > 0$.

Consequently $|A_d| = \varphi(d)$, including $d = 1$, when $A_1 = \{n\}$. Counting the partition gives

$$n = \sum_{d|n} |A_d| = \sum_{d|n} \varphi(d).$$

□

For example, the divisors of 12 are 1, 2, 3, 4, 6, 12, and

$$\varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12.$$

§6.5 Euler, Wilson and Fermat's Theorems

Euler–Fermat theorem (Euler's theorem):

Let $m \geq 2$ and let a be an integer with $\gcd(a, m) = 1$. Then

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

Proof: Choose a reduced residue system $r_1, \dots, r_{\varphi(m)}$ modulo m . Since a is coprime to m , the preceding result shows that $ar_1, \dots, ar_{\varphi(m)}$ is also a reduced residue system. Thus these two lists represent the same unit classes, possibly in a different order. Multiplying the corresponding congruences gives

$$(ar_1)(ar_2) \cdots (ar_{\varphi(m)}) \equiv r_1 r_2 \cdots r_{\varphi(m)} \pmod{m}.$$

Writing $R = r_1 r_2 \cdots r_{\varphi(m)}$, we obtain

$$a^{\varphi(m)} R \equiv R \pmod{m}.$$

Every r_i is coprime to m , so its class is a unit. Their product $[R]_m$ is therefore a unit as well, and $\gcd(R, m) = 1$. We may consequently cancel R from the congruence, yielding

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

□

Fermat's little theorem:

If p is prime, then $a^p \equiv a \pmod{p}$. This is a direct result of the above theorem.

For example, $\varphi(12) = 4$ and $\gcd(5, 12) = 1$, so $5^4 \equiv 1 \pmod{12}$. The coprimality assumption is essential: $\varphi(4) = 2$, but $2^2 \equiv 0 \not\equiv 1 \pmod{4}$.

Wilson's Theorem:

For every integer $n \geq 2$,

$$n \text{ is prime} \iff (n-1)! \equiv -1 \pmod{n}.$$

Proof for prime moduli: For $p = 2$, we have $(p-1)! = 1 \equiv -1 \pmod{2}$. Now let p be an odd prime. Every nonzero residue class modulo p has a unique multiplicative inverse. A class is its own inverse exactly when

$$a^2 \equiv 1 \pmod{p} \iff p \mid (a-1)(a+1).$$

By Euclid's lemma, this holds exactly when $a \equiv 1$ or $-1 \pmod{p}$. These two classes are distinct because p is odd.

Pair every remaining nonzero class with its inverse. Inversion applied twice returns the original class, so these are disjoint pairs, each with product 1. Multiplying all nonzero classes therefore gives

$$(p-1)! \equiv 1 \cdot (-1) \cdot 1 \cdots 1 \equiv -1 \pmod{p}.$$

Proof of the converse: Suppose $(n-1)! \equiv -1 \pmod{n}$, but n is composite. Then n has a prime divisor q with $2 \leq q < n$. Since q occurs among the factors of $(n-1)!$, we have $q \mid (n-1)!$. On the other hand, the assumed congruence gives $n \mid (n-1)! + 1$, so $q \mid (n-1)! + 1$ as well. Subtracting implies $q \mid 1$, a contradiction. Thus n is prime. $\square$

For example, $6! = 720 = 7 \cdot 103 - 1$, so $6! \equiv -1 \pmod{7}$. Unlike a Fermat congruence for a single base, Wilson's congruence is both necessary and sufficient for primality.

§6.6 Orders

For a unit $[a]_m$, its *order* $\text{ord}_m(a)$ is the least positive integer r such that $a^r \equiv 1 \pmod{m}$. Such an r exists by Euler's theorem.

The order divides the totient:

Let $n \geq 2$ and let a be an integer with $\gcd(a, n) = 1$. Then

$$\text{ord}_n(a) \mid \varphi(n).$$

Proof: Put $r = \text{ord}_n(a)$. By definition, $a^r \equiv 1 \pmod{n}$, and no smaller positive exponent has this property. Apply the division algorithm to $\varphi(n)$ with divisor r :

$$\varphi(n) = qr + s, \quad 0 \leq s < r.$$

Euler–Fermat gives $a^{\varphi(n)} \equiv 1 \pmod{n}$, so

$$1 \equiv a^{\varphi(n)} = a^{qr+s} = (a^r)^q a^s \equiv a^s \pmod{n}.$$

If $s > 0$, this contradicts the minimality of r , since $s < r$. Hence $s = 0$, and $\varphi(n) = qr$. Thus $r \mid \varphi(n)$. $\square$

More generally:

For every nonnegative integer t ,

$$a^t \equiv 1 \pmod{n} \iff \text{ord}_n(a) \mid t.$$

Indeed, if $a^t \equiv 1 \pmod{n}$, writing $t = qr + s$ with $0 \leq s < r$ and repeating the same argument forces $s = 0$. Conversely, if $t = qr$, then $a^t = (a^r)^q \equiv 1 \pmod{n}$. This includes $t = 0$.

In particular, if p is prime and $p \nmid a$, then $\text{ord}_p(a) \mid p - 1$, because $\varphi(p) = p - 1$.

The order need not equal the totient: modulo 7, we have $2^1 \equiv 2$, $2^2 \equiv 4$, and $2^3 \equiv 1$, so $\text{ord}_7(2) = 3$, a proper divisor of $\varphi(7) = 6$.

Theorem:

Let $n \geq 2$, let $\gcd(x, n) = 1$, and put $d = \text{ord}_n(x)$. For every nonnegative integer m ,

$$\text{ord}_n(x^m) = \frac{d}{\gcd(m, d)}.$$

Proof: If $m = 0$, then $x^m = 1$ has order 1, and the formula agrees because $\gcd(0, d) = d$. Now suppose $m > 0$. Write

$$g = \gcd(m, d), \quad m = gu, \quad d = gv, \quad \gcd(u, v) = 1.$$

For every positive integer t , the exponent criterion gives

$$\begin{aligned} (x^m)^t \equiv 1 \pmod{n} &\iff d \mid mt \\ &\iff v \mid ut \\ &\iff v \mid t. \end{aligned}$$

The last equivalence uses Euclid's lemma and $\gcd(u, v) = 1$. The least positive such t is therefore $v = d/g$, proving the formula. $\square$

For example, modulo 7, the successive powers of 3 are congruent to 3, 2, 6, 4, 5, 1, so $\text{ord}_7(3) = 6$. Hence

$$\text{ord}_7(3^4) = \frac{6}{\gcd(4, 6)} = 3.$$

§7 The Chinese Remainder Theorem

The Chinese remainder theorem combines congruences with different moduli into a single residue class. Coprime moduli allow arbitrary prescribed remainders, whereas common factors impose certain conditions in some cases.

§7.1 Two Coprime Moduli

Theorem:

Let $m, n \geq 2$ be coprime integers. For any integers a, b , the system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has a solution, unique modulo mn .

Proof of existence: By Bézout's identity, choose integers u, v such that $um + vn = 1$. Set

$$x_0 = avn + bum.$$

Modulo m , we have $vn \equiv 1$ and $um \equiv 0$, so $x_0 \equiv a$. Modulo n , we have $vn \equiv 0$ and $um \equiv 1$, so $x_0 \equiv b$.

Proof of uniqueness: If x, y are two solutions, then both m and n divide $x - y$. Since they are coprime, their product divides $x - y$. Conversely, adding any multiple of mn to a solution preserves both congruences. Thus all integer solutions are exactly

$$x = x_0 + mnt, \quad t \in \mathbb{Z}.$$

□

Uniqueness here means uniqueness of a residue class, not of an integer. In particular, exactly one solution lies in $\{0, 1, \dots, mn - 1\}$.

§7.2 Several Pairwise Coprime Moduli

Theorem:

Let $m_1, \dots, m_r \geq 2$ be pairwise coprime, meaning $\gcd(m_i, m_j) = 1$ whenever $i \neq j$. Put $M = m_1 \cdots m_r$. For arbitrary integers $a_1, \dots, a_r$, the system

$$x \equiv a_i \pmod{m_i} \quad (1 \leq i \leq r)$$

has exactly one solution modulo M .

Proof and construction: Put $M_i = M/m_i$. No prime divisor of m_i divides any other modulus, so $\gcd(M_i, m_i) = 1$. Hence there is an integer y_i with $M_i y_i \equiv 1 \pmod{m_i}$. Define

$$x_0 = \sum_{i=1}^r a_i M_i y_i.$$

Fix j . Every summand with $i \neq j$ is divisible by m_j , while the j th summand is congruent to a_j modulo m_j . Thus x_0 solves every congruence.

If x, y are solutions, each m_i divides $x - y$. Repeatedly applying divisibility by a product of coprime integers shows $M \mid x - y$: at each stage the product of the preceding moduli is coprime to the next modulus, since they have no common prime divisor. Conversely, every $x_0 + Mt$ satisfies the system. □

Example: Solve

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 2 \pmod{7}.$$

Here $M = 105$ and $(M_1, M_2, M_3) = (35, 21, 15)$. Their inverses modulo 3, 5, 7 are respectively 2, 1, 1, so

$$x_0 = 2 \cdot 35 \cdot 2 + 3 \cdot 21 + 2 \cdot 15 = 233 \equiv 23 \pmod{105}.$$

All solutions are $x = 23 + 105t$, $t \in \mathbb{Z}$.

§7.3 Residues and the Totient

For coprime $m, n \geq 2$, consider the reduction map

$$\Phi : \mathbb{Z}/mn\mathbb{Z} \longrightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}), \quad [x]_{mn} \longmapsto ([x]_m, [x]_n).$$

It is well-defined because $mn \mid x - y$ implies both $m \mid x - y$ and $n \mid x - y$. The existence and uniqueness parts of the theorem say precisely that Φ is a bijection. Reduction preserves addition, multiplication, and the identity, so this is an *isomorphism*: a bijection preserving these operations, with operations on pairs performed componentwise.

Correspondence of units:

The map restricts to a bijection

$$(\mathbb{Z}/mn\mathbb{Z})^* \longrightarrow (\mathbb{Z}/m\mathbb{Z})^* \times (\mathbb{Z}/n\mathbb{Z})^*.$$

Proof: An inverse modulo mn reduces to inverses modulo m and n . Conversely, suppose x has inverses u modulo m and v modulo n . By the Chinese remainder theorem, choose y with $y \equiv u \pmod{m}$ and $y \equiv v \pmod{n}$. Then $xy \equiv 1$ modulo both m and n , hence modulo mn . Thus $[x]_{mn}$ is a unit exactly when both its components are units. The stated bijection follows from the bijectivity of Φ . $\square$

Counting these finite sets provides another proof that

$$\varphi(mn) = \varphi(m)\varphi(n) \quad \text{if } \gcd(m, n) = 1.$$

If either modulus equals 1, this identity follows from $\varphi(1) = 1$. In terms of reduced residue systems, each pair consisting of a unit class modulo m and a unit class modulo n determines exactly one unit class modulo mn .

§7.4 Two Moduli That Need Not Be Coprime

Generalized Chinese remainder theorem for two moduli:

Let $m, n \geq 2$, put $g = \gcd(m, n)$, and let a, b be integers. The system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

is solvable if and only if

$$a \equiv b \pmod{g}.$$

When it is solvable, the solution is unique modulo $\text{lcm}(m, n) = mn/g$.

Proof: If x solves the system, g divides both $x - a$ and $x - b$, hence also $b - a$. This proves necessity.

Conversely, suppose $g \mid b - a$. Choose $u, v \in \mathbb{Z}$ with $um + vn = g$, and put $c = (b - a)/g$. Then

$$x_0 = a + muc$$

is congruent to a modulo m , and

$$x_0 - b = -(b - a) + muc = -gc + muc = -vnc$$

is divisible by n . Thus x_0 is a solution.

For uniqueness, if x, y are solutions, write $x - y = mt$. Since $n \mid mt$, canceling g gives $(n/g) \mid (m/g)t$. The integers $m/g, n/g$ are coprime, so Euclid's lemma implies $(n/g) \mid t$. Therefore $(mn/g) \mid x - y$. Conversely, mn/g is a multiple of both moduli, so adding any multiple of it preserves the system. This proves the claim. $\square$

For example, $x \equiv 1 \pmod{6}$ and $x \equiv 3 \pmod{8}$ are compatible because $1 \equiv 3 \pmod{2}$; their solutions are $x \equiv 19 \pmod{24}$. In contrast, $x \equiv 1 \pmod{6}$ and $x \equiv 2 \pmod{8}$ have no solution, since the first requires x to be odd and the second requires it to be even.

§7.5 The Generalized Theorem for Several Moduli

Generalized Chinese remainder theorem:

Let $m_1, \dots, m_r \geq 2$ be integers, not necessarily coprime, and let $a_1, \dots, a_r$ be integers. The system

$$x \equiv a_i \pmod{m_i} \quad (1 \leq i \leq r)$$

has a solution if and only if the remainders are pairwise compatible:

$$a_i \equiv a_j \pmod{\gcd(m_i, m_j)} \quad \text{for every } i, j.$$

When a solution exists, it is unique modulo

$$L = \text{lcm}(m_1, \dots, m_r).$$

Here the least common multiple of several positive integers is the smallest positive integer divisible by all of them.

Prime-power description of the lcm:

Let $\mathcal{P}$ be the finite set of primes dividing at least one modulus. By unique factorization, write

$$m_i = \prod_{p \in \mathcal{P}} p^{e_{i,p}}, \quad E_p = \max_{1 \leq i \leq r} e_{i,p},$$

where the exponents $e_{i,p}$ are nonnegative integers. Then

$$L = \prod_{p \in \mathcal{P}} p^{E_p}.$$

Indeed, this product is divisible by every m_i . Conversely, an integer divisible by every m_i is divisible by each p^{E_p} , since the maximum exponent occurs in some modulus. These prime powers are pairwise coprime, so their product divides that integer. Thus L divides every common multiple and is the least positive one.

Proof of necessity: If x is a solution, then $\gcd(m_i, m_j)$ divides both $x - a_i$ and $x - a_j$, and therefore divides their difference $a_j - a_i$. This gives every required compatibility condition.

Proof of sufficiency: Assume the pairwise compatibility conditions. For each $p \in \mathcal{P}$, choose an index $i(p)$ at which the maximum exponent occurs, so $e_{i(p),p} = E_p$. Consider the auxiliary system

$$x \equiv a_{i(p)} \pmod{p^{E_p}} \quad (p \in \mathcal{P}).$$

Its moduli are powers of distinct primes and hence pairwise coprime. The previously proved Chinese remainder theorem supplies a solution x_0 .

We check that x_0 solves the original system. Fix i and a prime p dividing m_i . Since $e_{i,p} \leq E_p$, the prime power $p^{e_{i,p}}$ divides both m_i and $m_{i(p)}$, hence their gcd. Pairwise compatibility therefore gives

$$a_i \equiv a_{i(p)} \pmod{p^{e_{i,p}}}.$$

The auxiliary system now implies $x_0 \equiv a_i \pmod{p^{e_{i,p}}}$. This holds for every prime power in the factorization of m_i . Since these prime powers are pairwise coprime, their product m_i divides $x_0 - a_i$. Thus every original congruence holds.

Proof of uniqueness: If x, y are solutions, every m_i divides $x - y$. By the lcm property proved above, $L \mid x - y$. Conversely, L is divisible by every modulus, so adding any multiple of L preserves all the congruences. Consequently all solutions are exactly

$$x = x_0 + Lt, \quad t \in \mathbb{Z}.$$

□

Example: Consider

$$x \equiv 1 \pmod{6}, \quad x \equiv 3 \pmod{8}, \quad x \equiv 4 \pmod{9}.$$

The pairwise gcds are 2, 3, 1, respectively, and the compatibility conditions hold: $1 \equiv 3 \pmod{2}$, $1 \equiv 4 \pmod{3}$, and any two integers are congruent modulo 1. The first two congruences give $x = 19 + 24t$. Substituting into the third yields

$$1 + 6t \equiv 4 \pmod{9} \iff 2t \equiv 1 \pmod{3} \iff t \equiv 2 \pmod{3}.$$

Hence all solutions are $x \equiv 67 \pmod{72}$, in agreement with $\text{lcm}(6, 8, 9) = 72$. The uniqueness modulus is the lcm, not generally the product of the moduli.

§8 Binomial Coefficients and Congruences

We collect the binomial coefficient identities and congruences here, including Lucas's theorem, Wolstenholme's theorem, and the binomial congruence used later to lift primitive roots to prime powers.

§8.1 Definitions and Basic Identities

For integers $N \geq 0$ and $0 \leq K \leq N$, define

$$\binom{N}{K} = \frac{N!}{K!(N-K)!}, \quad 0! = 1.$$

This is an integer: it counts the K -element subsets of an N -element set, since each such subset has $K!$ orderings, while there are $N(N-1) \cdots (N-K+1)$ ordered choices of K distinct elements. Set $\binom{N}{K} = 0$ for $K > N$, and in particular $\binom{0}{0} = 1$.

Basic identities:

The factorial formula gives

$$\binom{N}{K} = \binom{N}{N-K} \quad (0 \leq K \leq N), \quad K \binom{N}{K} = N \binom{N-1}{K-1} \quad (1 \leq K \leq N).$$

Pascal's identity is

$$\binom{N}{K} = \binom{N-1}{K} + \binom{N-1}{K-1} \quad (1 \leq K \leq N).$$

To prove it, fix one element of the N -element set and partition its K -element subsets according to whether they contain that element. The two counts on the right correspond to excluding and including it.

The binomial theorem:

For commuting variables X, Y and every integer $N \geq 0$,

$$(X + Y)^N = \sum_{K=0}^N \binom{N}{K} X^{N-K} Y^K.$$

Indeed, in expanding the product of N copies of $X + Y$, a term $X^{N-K} Y^K$ arises by choosing Y from exactly K of the factors. There are $\binom{N}{K}$ such choices. The case $N = 0$ is the identity $1 = 1$. $\square$

§8.2 Congruences of Fractions

To work with expressions such as p/k and harmonic sums, we extend congruence notation to rational numbers whose denominators are coprime to the modulus. If p is prime, $e \geq 1$, and $p \nmid BD$, define

$$\frac{A}{B} \equiv \frac{C}{D} \pmod{p^e} \iff p^e \mid AD - BC.$$

Equivalently, reduce the numerators modulo p^e and multiply by the inverses of the denominators.

This is independent of the chosen fractional representations: equality of fractions gives equality of their cross-products, and denominators are invertible modulo p^e . Addition and multiplication preserve these congruences, as follows from the usual residue-class arithmetic. In particular, for a fraction in lowest terms with denominator coprime to p , being congruent to 0 modulo p^e means that its numerator is divisible by p^e .

§8.3 Elementary Congruences for a Prime

Throughout this subsection, p is prime.

For every integer r with $1 \leq r \leq p-1$,

$$p \mid \binom{p}{r}.$$

Proof: The basic identity gives

$$r \binom{p}{r} = p \binom{p-1}{r-1}.$$

Since $p \nmid r$, Euclid's lemma implies $p \mid \binom{p}{r}$. The range matters: $\binom{p}{0} = \binom{p}{p} = 1$ is not divisible by p . $\square$

Polynomial congruence modulo a prime: For every prime p ,

$$(1 + T)^p \equiv 1 + T^p \pmod{p}.$$

Here congruence of integer polynomials means that corresponding coefficients are congruent modulo p .

Proof: By the binomial theorem,

$$(1 + T)^p = 1 + \sum_{r=1}^{p-1} \binom{p}{r} T^r + T^p.$$

The preceding remark gives $p \mid \binom{p}{r}$ for every $1 \leq r \leq p-1$. Thus every coefficient of $(1+T)^p - (1+T^p)$ is divisible by p , proving the congruence. $\square$

For $0 \leq r \leq p-1$,

$$\binom{p-1}{r} \equiv (-1)^r \pmod{p}.$$

Proof: The denominator $r!$ is invertible modulo p , so

$$\binom{p-1}{r} = \prod_{j=1}^r \frac{p-j}{j} \equiv \prod_{j=1}^r \frac{-j}{j} = (-1)^r \pmod{p}.$$

For $r = 0$, the products are empty and have value 1. $\square$

For $1 \leq k \leq p-1$,

$$\binom{p}{k} \equiv (-1)^{k-1} \frac{p}{k} \pmod{p^2}.$$

Here p/k means p times the inverse of k modulo p^2 .

Proof: The preceding congruence supplies an integer t such that

$$\binom{p-1}{k-1} = (-1)^{k-1} + pt.$$

Using $\binom{p}{k} = (p/k) \binom{p-1}{k-1}$ gives the exact rational identity

$$\binom{p}{k} - (-1)^{k-1} \frac{p}{k} = \frac{p^2 t}{k}.$$

As $p \nmid k$, the right side is congruent to zero modulo p^2 . Equivalently, the integer congruence $k \binom{p}{k} \equiv (-1)^{k-1} p \pmod{p^2}$ can be multiplied by the inverse of k . This proof also includes $p = 2$. $\square$

For example, with $p = 7$ and $k = 3$, the inverse of 3 modulo 49 is 33, so

$$\binom{7}{3} = 35 \equiv 7 \cdot 33 \equiv 35 \pmod{49}.$$

Binomial congruent to gif:

For every integer $k \geq 0$,

$$\binom{k}{p} \equiv \left\lfloor \frac{k}{p} \right\rfloor \pmod{p}.$$

Here $\lfloor x \rfloor$ is the greatest integer not exceeding x . The assertion includes $k < p$, when both sides are zero.

Proof: Write $k = qp + r$, where $0 \leq r < p$, we have:

$$(1 + T)^p \equiv 1 + T^p \pmod{p}.$$

Consequently,

$$(1 + T)^k \equiv (1 + T^p)^q (1 + T)^r \pmod{p}.$$

The coefficient of T^p on the left is $\binom{k}{p}$. On the right, since $r < p$, the only possible contribution is the T^p term from the first factor times the constant term from the second. Its coefficient is $\binom{q}{1} = q$, also zero when $q = 0$. Comparing coefficients proves the claim. $\square$

For example,

$$\binom{17}{5} = 6188 \equiv 3 = \left\lfloor \frac{17}{5} \right\rfloor \pmod{5}.$$

§8.4 A Binomial Congruence for Lifting

Lemma: If p is an odd prime, $s \geq 1$, and c is an integer, then

$$(1 + cp^s)^p \equiv 1 + cp^{s+1} \pmod{p^{s+2}}.$$

Proof: By the binomial theorem, the first two terms are 1 and cp^{s+1} . For $2 \leq j \leq p-1$, the prime-row divisibility lemma shows that $\binom{p}{j}$ is divisible by p . Thus the j th term is divisible by p^{1+sj} , and $1 + sj \geq s + 2$. The last term is $c^p p^{sp}$, and $sp \geq s + 2$ because $p \geq 3$. All remaining terms therefore vanish modulo p^{s+2} . $\square$

§8.5 Wolstenholme's Theorem: Harmonic Sums

Theorem:

Let $p \geq 5$ be prime. Then

$$H := \sum_{k=1}^{p-1} \frac{1}{k} \equiv 0 \pmod{p^2}, \quad S := \sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}.$$

Thus the numerator of H , when written in lowest terms, is divisible by p^2 , and the numerator of S is divisible by p . Their denominators are coprime to p , since a common denominator is

$$((p-1)!)^2.$$

Step 1: the reciprocal-square sum: Multiplication by 2 permutes the nonzero residue classes modulo p . Since the inverse square depends only on the nonzero residue class,

$$S = \sum_{k=1}^{p-1} \frac{1}{k^2} \equiv \sum_{k=1}^{p-1} \frac{1}{(2k)^2} = \frac{1}{4}S \pmod{p}.$$

Multiplying by 4 gives $3S \equiv 0 \pmod{p}$. As $p \geq 5$, the integer 3 is invertible modulo p , and hence $S \equiv 0 \pmod{p}$.

Step 2: the harmonic sum: The substitution $k \mapsto p-k$ permutes $1, \dots, p-1$, so

$$2H = \sum_{k=1}^{p-1} \left(\frac{1}{k} + \frac{1}{p-k} \right) = p \sum_{k=1}^{p-1} \frac{1}{k(p-k)}.$$

Modulo p , we have $k(p-k) \equiv -k^2$, so $1/(k(p-k)) \equiv -1/k^2$. Multiplying this congruence by p raises its modulus to p^2 . Summing therefore gives

$$2H \equiv -pS \equiv 0 \pmod{p^2},$$

where the last step uses Step 1. Since 2 is invertible modulo p^2 , we conclude that $H \equiv 0 \pmod{p^2}$.

□

For $p = 5$, for example,

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12},$$

whose numerator is divisible by 5^2 .

§8.6 Wolstenholme's Binomial Congruence

Wolstenholme's congruence:

For every prime $p \geq 5$,

$$\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}.$$

Equivalently, $\binom{2p}{p} \equiv 2 \pmod{p^3}$.

Proof: Write the binomial coefficient as the finite product

$$\binom{2p-1}{p-1} = \frac{(p+1)(p+2) \cdots (2p-1)}{(p-1)!} = \prod_{k=1}^{p-1} \left(1 + \frac{p}{k} \right).$$

On expanding, terms involving three or more factors p/k vanish modulo p^3 : each contains at least three factors of p , and its denominator is coprime to p . Hence

$$\binom{2p-1}{p-1} \equiv 1 + pH + p^2E \pmod{p^3}, \quad E = \sum_{1 \leq i < j \leq p-1} \frac{1}{ij}.$$

Expanding H^2 separates diagonal and off-diagonal terms:

$$H^2 = S + 2E.$$

The harmonic-sum theorem gives $H \equiv 0 \pmod{p^2}$ and $S \equiv 0 \pmod{p}$. Reducing this last identity modulo p and canceling 2 shows $E \equiv 0 \pmod{p}$. Thus $pH \equiv 0 \pmod{p^3}$ and $p^2E \equiv 0 \pmod{p^3}$, proving the desired congruence.

Finally, $\binom{2p}{p} = 2\binom{2p-1}{p-1}$. Multiplying by 2, or canceling its inverse modulo p^3 , proves the equivalence of the two binomial forms. $\square$

For example,

$$\binom{9}{4} = 126 = 1 + 5^3, \quad \binom{13}{6} = 1716 = 1 + 5 \cdot 7^3.$$

The restriction $p \geq 5$ matters: for $p = 3$, $\binom{5}{2} = 10 \not\equiv 1 \pmod{27}$, and for $p = 2$, $\binom{3}{1} = 3 \not\equiv 1 \pmod{8}$.

§9 Primitive Roots

§9.1 Definition and Powers of a Primitive Root

We finally go back to orders of a group, and contemplate if every group must have an element g such that $\text{ord}_m(g) = \phi(m)$

Let $n \geq 2$. An integer g is a *primitive root modulo n* if

$$\gcd(g, n) = 1 \quad \text{and} \quad \text{ord}_n(g) = \varphi(n).$$

A group G is *cyclic* if some element $u \in G$ generates it:

$$G = \langle u \rangle = \{u^j : j \in \mathbb{Z}\}.$$

Here negative powers are defined using the inverse of u . Such an element u is called a *generator*.

Theorem:

For every integer $n \geq 2$,

$$(\mathbb{Z}/n\mathbb{Z})^* \text{ is cyclic} \iff n \text{ has a primitive root.}$$

More precisely, a unit class $[g]_n$ generates the unit group if and only if its integer representative g is a primitive root modulo n .

Proof: First let g be any integer coprime to n , and put $d = \text{ord}_n(g)$. Its powers

$$[1]_n, [g]_n, \dots, [g^{d-1}]_n$$

are distinct. Indeed, if $g^i \equiv g^j \pmod{n}$ with $0 \leq i < j < d$, canceling the unit $[g^i]_n$ gives $g^{j-i} \equiv 1 \pmod{n}$, contrary to the minimality of d . Every integer exponent t , including a negative one, can be written $t = qd + s$ with $0 \leq s < d$. Since $[g]_n^d = [1]_n$, we have $[g]_n^t = [g]_n^s$. Thus

$$\langle [g]_n \rangle = \{[1]_n, [g]_n, \dots, [g^{d-1}]_n\}, \quad |\langle [g]_n \rangle| = d.$$

Primitive root implies cyclicity: If g is a primitive root modulo n , then $d = \varphi(n)$. Its powers therefore form a subset of the unit group with exactly $\varphi(n)$ elements. Since the whole unit group also has $\varphi(n)$ elements, the subset is the entire group. Hence $(\mathbb{Z}/n\mathbb{Z})^* = \langle [g]_n \rangle$ is cyclic.

Cyclicity implies a primitive root: Conversely, suppose the unit group is cyclic, generated by a class $[g]_n$. This class is a unit, so $\gcd(g, n) = 1$. As its powers comprise the whole unit group, the count above gives

$$\text{ord}_n(g) = |\langle [g]_n \rangle| = |(\mathbb{Z}/n\mathbb{Z})^*| = \varphi(n).$$

Thus g is a primitive root modulo n . □

The group element is the residue class $[g]_n$, whereas a primitive root is an integer representative of that class. Replacing g by a congruent integer does not change its powers modulo n or its order.

For $n = 2$, the unit group is the one-element cyclic group $\{[1]_2\}$, and 1 is a primitive root, so the equivalence includes this case.

For example, 3 is a primitive root modulo 7, since its order is $6 = \varphi(7)$, whereas 2 is not, since its order is 3.

Powers that remain primitive:

If g is a primitive root modulo n and $m \geq 0$, then

$$g^m \text{ is a primitive root modulo } n \iff \gcd(m, \varphi(n)) = 1.$$

Indeed, the order-of-a-power formula gives

$$\text{ord}_n(g^m) = \frac{\varphi(n)}{\gcd(m, \varphi(n))},$$

which equals $\varphi(n)$ exactly under the stated condition.

§9.2 An Obstruction from Coprime Factors

Lemma: For every integer $k > 2$, the number $\varphi(k)$ is even.

Proof: Pair each unit class $[u]_k$ with $[-u]_k$. Negation takes units to units and applying it twice returns the original class. There are no fixed points: if $u \equiv -u \pmod{k}$, then $2u \equiv 0 \pmod{k}$, and cancellation of the unit u gives $k \mid 2$, contrary to $k > 2$. Thus the unit classes split into disjoint pairs, so their number $\varphi(k)$ is even. □

Theorem:

If $n = ab$ with $a > 2$, $b > 2$, and $\gcd(a, b) = 1$, then n has no primitive root.

Proof: Let x be any integer coprime to n . It is coprime to both a and b . Put

$$L = \text{lcm}(\varphi(a), \varphi(b)).$$

Euler–Fermat gives $x^{\varphi(a)} \equiv 1 \pmod{a}$ and $x^{\varphi(b)} \equiv 1 \pmod{b}$. Since both exponents divide L , we have $x^L \equiv 1$ modulo both a and b . Their coprimality implies $ab \mid x^L - 1$, so

$$\text{ord}_n(x) \mid L.$$

By the lemma, $\varphi(a)$ and $\varphi(b)$ are both even. The gcd-lcm identity and multiplicativity of the totient therefore give

$$L = \frac{\varphi(a)\varphi(b)}{\gcd(\varphi(a), \varphi(b))} \leq \frac{\varphi(a)\varphi(b)}{2} = \frac{\varphi(n)}{2} < \varphi(n).$$

Consequently every unit has order strictly smaller than $\varphi(n)$, so no primitive root exists. $\square$

For example, $15 = 3 \cdot 5$ has no primitive root: every unit has order dividing $\text{lcm}(2, 4) = 4$, while $\varphi(15) = 8$.

Why the factor restrictions matter: Coprimality of a, b alone is not enough. The integer $6 = 2 \cdot 3$ has primitive root 5, since $5 \not\equiv 1 \pmod{6}$ but $5^2 \equiv 1 \pmod{6}$, so $\text{ord}_6(5) = 2 = \varphi(6)$. Allowing a factor 1 would also make the claim false. Thus the obstruction proved here requires both coprime factors to exceed 2.

§9.3 Classification: The Necessary Forms

A very natural question upon knowing about primitive roots, would be to ask which numbers can have primitive roots?

Primitive-root classification:

An integer $N \geq 2$ has a primitive root if and only if

$$N = 2, \quad 4, \quad p^k, \quad \text{or} \quad 2p^k,$$

where p is an odd prime and $k \geq 1$. We first prove that no other moduli are possible, then prove existence for every modulus on this list. Our definition of primitive roots uses moduli at least 2, so modulus 1 is not included.

Lemma: higher powers of two have no primitive roots: If $s \geq 3$ and x is odd, then

$$x^{2^{s-2}} \equiv 1 \pmod{2^s}.$$

Proof: For $s = 3$, write $x = 2t + 1$. Then $x^2 - 1 = 4t(t + 1)$ is divisible by 8, since one of $t, t + 1$ is even. Suppose the congruence holds for s , so $x^{2^{s-2}} = 1 + c2^s$ for some integer c . Squaring gives

$$x^{2^{s-1}} = (1 + c2^s)^2 = 1 + c2^{s+1} + c^2 2^{2s} \equiv 1 \pmod{2^{s+1}}.$$

Induction proves the congruence for every $s \geq 3$. Thus every unit modulo 2^s has order at most 2^{s-2} , strictly less than $\varphi(2^s) = 2^{s-1}$. No primitive root exists. $\square$

Proof of necessity in the classification: Suppose $N \geq 2$ has a primitive root, and write its prime factorization as

$$N = 2^s p_1^{e_1} \cdots p_r^{e_r},$$

where $s \geq 0$, the p_i are distinct odd primes, and $e_i \geq 1$. If $r \geq 2$, take $a = p_1^{e_1}$ and $b = N/a$. Then $a, b > 2$ and $\gcd(a, b) = 1$, contrary to the coprime-factor obstruction. Hence $r \leq 1$.

If $r = 1$ and $s \geq 2$, the factorization $N = 2^s p_1^{e_1}$ again has two coprime factors exceeding 2. Thus in this case $s = 0$ or 1 , giving $N = p_1^{e_1}$ or $2p_1^{e_1}$. If $r = 0$, then $N = 2^s$ with $s \geq 1$. The preceding lemma excludes $s \geq 3$, leaving $N = 2$ or 4 . This proves the necessary direction. $\square$

§9.4 Existence of Primitive Roots Modulo a Prime

Polynomial root bound:

Let p be prime. A polynomial of degree $d \geq 0$ with coefficients in $\mathbb{Z}/p\mathbb{Z}$ and nonzero leading coefficient has at most d distinct roots modulo p .

Proof: We induct on d . A nonzero constant has no roots. For $d \geq 1$, there is nothing to prove if the polynomial f has no root. Otherwise choose a root a . The identities

$$T^j - a^j = (T - a)(T^{j-1} + aT^{j-2} + \cdots + a^{j-1}) \quad (j \geq 1)$$

show that $f(T) - f(a) = (T - a)h(T)$ for a polynomial h of degree $d - 1$ with the same nonzero leading coefficient as f . Since $f(a) = 0$, any other root $b \neq a$ satisfies $0 = (b - a)h(b)$. The nonzero class $b - a$ is invertible modulo p , so $h(b) = 0$. Induction bounds these other roots by $d - 1$, giving at most d roots in total. $\square$

Theorem:

Every prime p has a primitive root.

Proof: For each positive divisor d of $p - 1$, let $A(d)$ be the number of nonzero residue classes modulo p having order exactly d . Every nonzero class has order dividing $p - 1$, so

$$\sum_{d|p-1} A(d) = p - 1.$$

We claim that $A(d)$ is either 0 or $\varphi(d)$. If $A(d) > 0$, choose a class a of order d . Its d distinct powers $1, a, \dots, a^{d-1}$ are roots of $T^d - 1$. By the root bound these are all the roots. Every class of order d is one of them, and the order-of-a-power formula shows that a^j has order d exactly when $\gcd(j, d) = 1$. There are $\varphi(d)$ such exponents in $0 \leq j < d$. This includes $d = 1$, where the sole exponent is 0 and $\gcd(0, 1) = 1$. Thus the claim follows.

In particular, $A(d) \leq \varphi(d)$ for every $d | p - 1$. But the divisor-sum identity gives

$$\sum_{d|p-1} A(d) = p - 1 = \sum_{d|p-1} \varphi(d).$$

Since each difference $\varphi(d) - A(d)$ is nonnegative and their sum is zero, every difference is zero. In particular, $A(p - 1) = \varphi(p - 1) > 0$. Hence a class of order $p - 1$ exists, which is a primitive root modulo p . $\square$

§9.5 Lifting to Powers of an Odd Prime

Lifting criterion for a given primitive root:

Let p be an odd prime, and suppose r is a primitive root modulo p such that

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

Then the same integer r is a primitive root modulo p^n for every integer $n \geq 1$. More precisely,

$$\text{ord}_{p^n}(r) = (p - 1)p^{n-1}.$$

Proof: Since r is a primitive root modulo p , we have $p \nmid r$ and $r^{p-1} \equiv 1 \pmod{p}$. The extra hypothesis therefore means

$$r^{p-1} = 1 + c_0 p, \quad p \nmid c_0.$$

We first prove by induction on $j \geq 0$ that

$$r^{(p-1)p^j} = 1 + c_j p^{j+1}, \quad p \nmid c_j.$$

The base case was just established. If the identity holds for j , raise it to the p th power. The binomial congruence from Subsection 8.4, with $s = j + 1$, gives

$$r^{(p-1)p^{j+1}} \equiv 1 + c_j p^{j+2} \pmod{p^{j+3}}.$$

Thus $r^{(p-1)p^{j+1}} - 1$ is divisible by p^{j+2} , and its quotient c_{j+1} is congruent to c_j modulo p . In particular, $p \nmid c_{j+1}$, completing the induction.

Now fix $n \geq 2$ and put $D = \text{ord}_{p^n}(r)$. Reducing $r^D \equiv 1 \pmod{p^n}$ modulo p shows $\text{ord}_p(r) = p - 1 \mid D$. Also, the order divides the totient:

$$D \mid \varphi(p^n) = (p - 1)p^{n-1}.$$

Writing $D = (p - 1)e$ and canceling $p - 1$ in this divisibility shows $e \mid p^{n-1}$. Hence

$$D = (p - 1)p^t \quad \text{for some } 0 \leq t \leq n - 1.$$

If $t < n - 1$, the induction formula says

$$r^D - 1 = c_t p^{t+1}, \quad p \nmid c_t.$$

Since $t + 1 < n$, this is not divisible by p^n , contradicting the definition of D . Therefore $t = n - 1$, and $D = (p - 1)p^{n-1} = \varphi(p^n)$. The case $n = 1$ is the hypothesis that r is primitive modulo p . $\square$

The extra hypothesis is necessary for lifting to p^2 : If instead $r^{p-1} \equiv 1 \pmod{p^2}$, then $\text{ord}_{p^2}(r) \mid p - 1$, so r cannot have order $\varphi(p^2) = p(p - 1)$.

For instance, 8 is a primitive root modulo 3, but $8^2 = 64 \equiv 1 \pmod{9}$, so it is not primitive modulo 9. In contrast, 2 is primitive modulo 3 and $2^2 = 4 \not\equiv 1 \pmod{9}$; the criterion shows that 2 is primitive modulo every 3^n .

Corollary: a suitable lift always exists:

For every odd prime p , there is an integer h that is a primitive root modulo p^k for every $k \geq 1$.

Proof: Choose an integer g that is a primitive root modulo p . If $g^{p-1} \not\equiv 1 \pmod{p^2}$, take $h = g$. Otherwise, the binomial theorem gives

$$(g + p)^{p-1} \equiv g^{p-1} + (p - 1)g^{p-2}p \equiv 1 + (p - 1)g^{p-2}p \pmod{p^2}.$$

Since p divides neither $p - 1$ nor g , the coefficient $(p - 1)g^{p-2}$ is nonzero modulo p . Thus take $h = g + p$. In either case, $h \equiv g \pmod{p}$ is still a primitive root modulo p , and $h^{p-1} \not\equiv 1 \pmod{p^2}$. The lifting criterion now applies. $\square$

§9.6 Completing the Classification

Modulo 2, the class of 1 has order $1 = \varphi(2)$. Modulo 4, the class of 3 has order $2 = \varphi(4)$. Thus both moduli have primitive roots.

Now let p be an odd prime and $k \geq 1$. The preceding corollary supplies a primitive root h modulo p^k . Choose an odd representative of this class: take $g = h$ if h is odd, and $g = h + p^k$ otherwise.

Then g is coprime to $2p^k$ and still has order $\varphi(p^k)$ modulo p^k . For any positive integer t , oddness gives $g^t \equiv 1 \pmod{2}$, so coprime-product divisibility yields

$$g^t \equiv 1 \pmod{2p^k} \iff g^t \equiv 1 \pmod{p^k}.$$

The least positive exponents in these two congruences are therefore equal. Since $\varphi(2p^k) = \varphi(2)\varphi(p^k) = \varphi(p^k)$, we obtain

$$\text{ord}_{2p^k}(g) = \varphi(2p^k).$$

Thus $2p^k$ also has a primitive root. Together with the necessary direction, this proves the full classification. $\square$

§9.7 Artin's Primitive Root Conjecture

Artin's conjecture (infinitude form):

Let a be a fixed integer that is neither -1 nor a perfect square. Then there are infinitely many primes p for which a is a primitive root modulo p ; equivalently,

$$\#\{p : p \text{ is prime, } p \nmid a, \text{ord}_p(a) = p - 1\} = \infty.$$

For example, the conjecture predicts that 2 is a primitive root modulo infinitely many primes.

§10 Primes in Arithmetic Progressions

We prove that each of the forms

$$4k + 1, \quad 8k + 1, \quad 2^l k + 1, \quad 4k + 3, \quad 8k + 3$$

contains infinitely many primes, with $k \geq 0$ an integer. These proofs use finite constructions and the congruence tools already established.

§10.1 Primes Congruent to 1 Modulo 4

Suppose all primes congruent to 1 modulo 4 formed a finite list, and let P be their product. Set

$$A = 2P, \quad N = A^2 + 1.$$

Any prime divisor q of N is odd, and $q \nmid A$. Modulo q ,

$$A^2 \equiv -1, \quad A^4 \equiv 1.$$

Since $-1 \not\equiv 1 \pmod{q}$, the order of A modulo q divides 4 but does not divide 2, so it is exactly 4. The order divides $q - 1$, hence $q \equiv 1 \pmod{4}$. But every listed prime divides A and therefore cannot divide $A^2 + 1$. Thus q is a new prime of the required form, a contradiction. $\square$

§10.2 Primes Congruent to 1 Modulo 8

Suppose there were only finitely many such primes, and let P be their product. Put

$$A = 2P, \quad N = A^4 + 1.$$

For any prime divisor q of N , the prime q is odd and

$$A^4 \equiv -1 \pmod{q}, \quad A^8 \equiv 1 \pmod{q}.$$

The order of A modulo q divides 8 but not 4. Since the proper positive divisors of 8 all divide 4, that order is 8. Hence $8 \mid q - 1$, so $q \equiv 1 \pmod{8}$. No listed prime can divide N , because each divides A . This contradiction proves the claim. $\square$

§10.3 Primes Congruent to 1 Modulo a Power of 2

Theorem:

For every fixed integer $\ell \geq 0$, there are infinitely many primes of the form

$$p = 2^\ell x + 1, \quad x \in \mathbb{Z}_{>0}.$$

Equivalently, infinitely many primes satisfy $p \equiv 1 \pmod{2^\ell}$. The exponent ℓ is fixed; the positive integer x varies.

Proof: For $\ell = 0$, every prime has the form $x + 1$ with $x = p - 1 > 0$, so the assertion is the infinitude of primes. Now fix $\ell \geq 1$. Suppose, for a contradiction, that all primes congruent to 1 modulo 2^ℓ form a finite list $p_1, \dots, p_r$. Put

$$P = p_1 \cdots p_r, \quad A = 2P, \quad N = A^{2^{\ell-1}} + 1,$$

where $P = 1$ if the list is empty. Since A is even and positive, $N > 1$ is odd. Choose a prime divisor q of N . Then q is odd and $q \nmid A$, since a common divisor of A and N would divide 1. Modulo q , we have

$$A^{2^{\ell-1}} \equiv -1, \quad A^{2^\ell} \equiv 1.$$

Let $d = \text{ord}_q(A)$. The second congruence implies $d \mid 2^\ell$. The first implies $d \nmid 2^{\ell-1}$, because $-1 \not\equiv 1 \pmod{q}$. Every proper positive divisor of 2^ℓ divides $2^{\ell-1}$, so necessarily $d = 2^\ell$. The order divides $q - 1$, and hence

$$2^\ell \mid q - 1, \quad q = 2^\ell x + 1 \quad \text{with } x = \frac{q - 1}{2^\ell} > 0.$$

Thus q has the required form. But each listed prime divides A , so $N \equiv 1 \pmod{p_i}$ for every i . None can equal q , contradicting the completeness of the list. Therefore there are infinitely many such primes. $\square$

Taking $\ell = 2$ and $\ell = 3$ gives the classes $4x + 1$ and $8x + 1$, respectively. The proofs below spell out these special cases.

§10.4 Primes Congruent to 3 Modulo 4

Suppose the primes congruent to 3 modulo 4 formed a finite list, and let P be their product. The integer

$$N = 4P - 1 > 1$$

is odd and congruent to 3 modulo 4. Every prime factor of N is congruent to 1 or 3 modulo 4. If all were congruent to 1, their product, with multiplicities, would also be congruent to 1. Therefore N has a prime divisor $q \equiv 3 \pmod{4}$. None of the listed primes divides N , since $N \equiv -1$ modulo each of them. Again we obtain a prime outside the supposedly complete list. $\square$

§10.5 Primes Congruent to 3 Modulo 8

We first establish the necessary restriction on prime divisors of integers of the form $t^2 + 2$.

Lemma: If q is an odd prime and $t^2 \equiv -2 \pmod{q}$ for an integer t , then $q \equiv 1$ or $3 \pmod{8}$.

Proof: Put $h = (q - 1)/2$. In the list $2, 4, \dots, 2h$, replace each entry larger than $q/2$ by that entry minus q . The absolute values of the resulting signed integers are a permutation of $1, \dots, h$. Indeed,

equality of two absolute values would give $2j \equiv \pm 2k \pmod{q}$. Cancellation gives $j = k$ in the plus case, while the minus case is impossible because $2 \leq j + k \leq q - 1$. There are exactly

$$\nu = h - \left\lfloor \frac{q}{4} \right\rfloor$$

negative entries. Multiplying all entries and canceling the invertible factor $h!$ yields

$$2^h \equiv (-1)^\nu \pmod{q}, \quad (-2)^h \equiv (-1)^{\lfloor q/4 \rfloor} \pmod{q}.$$

Since $q \nmid t$, Fermat's little theorem gives $(-2)^h \equiv t^{q-1} \equiv 1 \pmod{q}$. Thus $\lfloor q/4 \rfloor$ is even. Among the four odd residue classes 1, 3, 5, 7 modulo 8, this holds exactly for 1 and 3. $\square$

Now suppose all primes congruent to 3 modulo 8 were in a finite list, with product P . This product is odd. Every odd square is congruent to 1 modulo 8, since $(2r+1)^2 = 4r(r+1) + 1$ and $r(r+1)$ is even. Hence

$$N = P^2 + 2 \equiv 3 \pmod{8}.$$

Every prime divisor q of N is odd and satisfies $P^2 \equiv -2 \pmod{q}$, so the lemma puts it in class 1 or 3 modulo 8. Not all prime factors of N can be in class 1, since their product would then be 1 modulo 8. Thus some divisor q is 3 modulo 8. But no listed prime divides N : each divides P , and would therefore have to divide 2. This gives the required contradiction. $\square$

§11 Arithmetic Functions

Definition: An arithmetic function is a function from $\mathbb{N}$ to $\mathbb{C}$. Denote the set of all arithmetic functions by $\mathcal{A}$.

Definition: An arithmetic function is called *multiplicative* if

$$f(mn) = f(m)f(n)$$

for all coprime m, n . It is called *completely multiplicative* if the property holds for all m, n .

Definition: An arithmetic function is called *additive* if

$$f(mn) = f(m) + f(n)$$

for all coprime m, n . It is called *completely additive* if the property holds for all m, n .

§11.1 Number of Divisors

Definition: The number of divisors function $d(n)$ is defined as

$$d(n) := \sum_{d|n} 1$$

Theorem:

For natural number $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, we have

$$d(n) = (1 + \alpha_1) \dots (1 + \alpha_k)$$

Proof: Each divisor can be written as $p_1^{e_1} \dots p_k^{e_k}$ where $e_i \in \{0, 1, \dots, \alpha_i\}$. Since there are $\alpha_i + 1$ many choices for each exponent, simply multiplying them gives us the number of divisors possible. $\square$

Lemma: $d(n)$ is odd $\iff n$ is a perfect square

Proof: ($\Rightarrow$) Note that $d(n)$ is a product of numbers, so if it wants to be odd, all of the factors getting multiplied must be odd.

$$\Rightarrow \alpha_i + 1 = 2k_i + 1$$

$$\Rightarrow \alpha_i = 2k_i$$

Therefore, every exponent is even, and thus n must be a perfect square. ($\Leftarrow$) n is a perfect square, so all the α_i are even, which means all $\alpha_i + 1$ are odd, and therefore $d(n)$ is odd. $\square$

Lemma: $d(n) \leq 2\sqrt{n}$

Proof: There exists a divisor $d \leq \sqrt{n}$ (say 1) and equivalently, there exists another divisor $d' = n/d$ such that $d' \geq \sqrt{n}$ (say n). So, there exist two divisors on either side of $\sqrt{n}$. Thus, there will be as many divisors from 1 to $\sqrt{n}$ as there will be divisors from $\sqrt{n}$ to n . The number of divisors from 1 to $\sqrt{n}$ can be at most $\sqrt{n}$, and thus there can be at most $\sqrt{n}$ many divisors from $\sqrt{n}$ to n as well. Therefore we have,

$$d(n) \leq \sqrt{n} + \sqrt{n} = 2\sqrt{n}$$

$\square$

Note: $d(n)$ is neither multiplicative nor additive.

§11.2 Sum of Divisors

The sum of divisors function $\sigma(n)$ is defined as

$$\sigma(n) := \sum_{d|n} d$$

The generalised sum of divisors function $\sigma_k(n)$ is defined as

$$\sigma_k(n) := \sum_{d|n} d^k$$

Theorem:

For natural number $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, we have

$$\sigma(n) = \prod_{i=1}^k \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}$$

Proof:

$$\begin{aligned} \sigma(n) &:= \sum_{d|n} d \\ \Rightarrow \sigma(n) &= \sum_{e_i \leq \alpha_i} p_1^{e_1} \dots p_k^{e_k} \\ &= \sum_{e_i \leq \alpha_i} \prod_{j=1}^k p_j^{e_j} \\ &= \prod_{i=1}^k \sum_{j=1}^{\alpha_i} p_i^j \\ &= \prod_{i=1}^k \frac{p_i^{\alpha_i+1} - 1}{p_i - 1} \end{aligned}$$

□

Note: This implies that $\sigma(n)$ is multiplicative but not completely multiplicative.

§11.3 Euler Totient Function

The Euler Totient function $\phi(n)$ is defined as the number of coprime natural numbers less than n , or

$$\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^*|$$

The following results regarding $\phi(n)$ have already been proved in earlier sections.

Formula:

For every integer $m \geq 2$,

$$\varphi(m) = m \prod_{p|m} \left(1 - \frac{1}{p}\right),$$

where the product is over the distinct prime divisors of m .

Multiplicativity:

If m, n are positive integers with $\gcd(m, n) = 1$, then

$$\varphi(mn) = \varphi(m)\varphi(n).$$

Divisor-sum identity:

For every positive integer n ,

$$\sum_{d|n} \varphi(d) = n,$$

where the sum is over the positive divisors of n .

Note: $\phi(n)$ is multiplicative but not completely multiplicative.

§11.4 Some Other Arithmetic Functions**Möbius function**

We define the Möbius function as:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 \dots p_k \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Dirichlet Delta function

We define the Dirichlet Delta function as:

$$\delta(n) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases} \quad (2)$$

Note: Both the Möbius function and Dirichlet Delta function are multiplicative.

Others: For $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$

$$id(n) = n \quad (3)$$

$$w(n) = k \quad (4)$$

$$\Omega(n) = p_1^{\alpha_1} + \dots + p_k^{\alpha_k} \quad (5)$$

§11.5 Dirichlet Convolution**Dirichlet Convolution**

Let $f, g \in \mathcal{A}$ be two such functions. We define their *Dirichlet Convolution* as such:

$$f * g(n) := \sum_{d|n} f(d)g(n/d)$$

For example,

$$f * g(6) = f(1)g(6) + f(2)g(3) + f(3)g(2) + f(6)g(1)$$

Properties: The following statements hold:

1. $f * g = g * f$

2. $(f * g) * h = f * (g * h)$
3. $f * (g + h) = f * g + f * h$
4. $f * \delta = f$ (Surprisingly the identity for $*$ turns out to be δ rather than id or 1).
5. If f, g are multiplicative, so is $f * g$
6. If $f, f * g$ are multiplicative, so is g

This gives that $(f, *)$ is an abelian group. The proof is left as an exercise to the reader.

Lemma: The following statements hold:

1. $\mu * 1 = \delta$
2. $1 * 1 = d$
3. $id * 1 = \sigma$
4. $\phi * 1 = id$

The proof is left as an exercise to the reader. An interesting result that easily follows is that $\phi * d = \sigma$.

§11.6 Möbius Inversion

We know that δ is the identity for convolution, so it makes sense to define the inverse of a function f as a function g such that $f * g = \delta$.

Möbius Inversion

Let $f, g \in \mathcal{A}$. Then for all $n \in \mathbb{N}$,

$$F = f * 1 \iff f = \mu * F$$

Proof: $(\Rightarrow) F = f * 1$

$$\Rightarrow F * \mu = (f * 1) * \mu = f * (1 * \mu) = f * \delta = f$$

$$\Rightarrow F * \mu = f$$

$(\Leftarrow) f = F * \mu$

$$\Rightarrow f * 1 = (F * \mu) * 1 = F * (\mu * 1) = F * \delta = F$$

$$\Rightarrow f * 1 = F$$

□

Theorem: If $f \in \mathcal{A}$, then f^{-1} exists $\iff f(1) \neq 0$

Proof: $(\Rightarrow) f^{-1}$ exists

$$\Rightarrow (f * f^{-1})(1) = 1$$

$$\Rightarrow f(1)f^{-1}(1) = 1$$

$$\Rightarrow f(1) \neq 0$$

Proof: $(\Leftarrow) f(1) \neq 0$

$$\Rightarrow f^{-1}(1) := \frac{1}{f(1)}$$

Thus we know $f^{-1}(1)$. Inductively, assume we know $f^{-1}(2), f^{-1}(3), \dots, f^{-1}(n-1)$.

$$\Rightarrow (f * f^{-1})(n) = 0$$

$$\Rightarrow \sum_{d|n} f(d) f^{-1}\left(\frac{n}{d}\right) = 0$$

$$\Rightarrow f(1) f^{-1}(n) + \sum_{\substack{d|n \\ d>1}} f(d) f^{-1}\left(\frac{n}{d}\right) = 0$$

$$\Rightarrow f^{-1}(n) = \frac{-1}{f(1)} \sum_{\substack{d|n \\ d>1}} f(d) f^{-1}\left(\frac{n}{d}\right)$$

Thus $f^{-1}(n)$ is known. So, by induction, f^{-1} exists.

§11.7 The Von Mangoldt Function

We define the Von Mangoldt function as:

$$\Lambda(n) = \begin{cases} \log p & n = p^m \text{ for some } p \text{ prime and } m \text{ natural number} \\ 0 & \text{otherwise} \end{cases}$$

Von Mangoldt divisor-sum identity:

For every positive integer n ,

$$\sum_{d|n} \Lambda(d) = \log n.$$

Equivalently, $\Lambda * 1 = \log$, where $\log$ denotes the arithmetic function $n \mapsto \log n$.

Proof: For $n = 1$, both sides are zero. Now let $n > 1$ and write its prime factorization as

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i \geq 1$. Among the positive divisors of n , the only ones on which Λ is nonzero are the prime powers

$$p_i, p_i^2, \dots, p_i^{\alpha_i} \quad (1 \leq i \leq k).$$

Each of these contributes $\log p_i$. Therefore

$$\sum_{d|n} \Lambda(d) = \sum_{i=1}^k \sum_{j=1}^{\alpha_i} \Lambda(p_i^j) = \sum_{i=1}^k \alpha_i \log p_i = \log \left(\prod_{i=1}^k p_i^{\alpha_i} \right) = \log n.$$

□

Corollary: $\Lambda(n) = \mu * \log = \sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) = - \sum_{d|n} \mu(d) \log(d)$

§11.8 Liouville's Function

We define Liouville's function by

$$\lambda(1) = 1, \quad \lambda(n) = (-1)^{\alpha_1 + \dots + \alpha_k}$$

when

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$$

Liouville divisor-sum identity:

For every positive integer n ,

$$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n \text{ is a perfect square,} \\ 0 & \text{otherwise.} \end{cases}$$

Proof: Write

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i \geq 1$. Every positive divisor of n has the form

$$d = p_1^{e_1} \cdots p_k^{e_k}, \quad 0 \leq e_i \leq \alpha_i,$$

and hence $\lambda(d) = (-1)^{e_1 + \dots + e_k}$. Therefore

$$\sum_{d|n} \lambda(d) = \prod_{i=1}^k \left(\sum_{e_i=0}^{\alpha_i} (-1)^{e_i} \right).$$

For each i ,

$$\sum_{e_i=0}^{\alpha_i} (-1)^{e_i} = \begin{cases} 1 & \text{if } \alpha_i \text{ is even,} \\ 0 & \text{if } \alpha_i \text{ is odd.} \end{cases}$$

Thus the product equals 1 exactly when every α_i is even, which is equivalent to n being a perfect square; otherwise it equals 0. The case $n = 1$ also gives $\lambda(1) = 1$. $\square$

Dirichlet inverse of Liouville's function:

The Dirichlet inverse of λ is the absolute value of the Möbius function:

$$\lambda^{-1} = |\mu|.$$

Equivalently,

$$\lambda * |\mu| = \delta.$$

Proof: Both λ and $|\mu|$ are multiplicative, so their Dirichlet convolution is multiplicative. It

is therefore enough to evaluate the convolution at prime powers. For every prime p ,

$$\lambda(p^j) = (-1)^j, \quad |\mu(p^j)| = \begin{cases} 1 & \text{if } j = 0 \text{ or } j = 1, \\ 0 & \text{if } j \geq 2. \end{cases}$$

$$(\lambda * |\mu|)(p^a) = \lambda(p^a)|\mu(1)| + \lambda(p^{a-1})|\mu(p)| + 0 = (-1)^a + (-1)^{a-1} = 0.$$

At $n = 1$ we have

$$(\lambda * |\mu|)(1) = \lambda(1)|\mu(1)| = 1.$$

Thus $\lambda * |\mu|$ equals 1 at 1 and 0 at every prime power of positive exponent.

$$\Rightarrow \lambda * |\mu| = \delta$$

Therefore $\lambda^{-1} = |\mu|$. □

§12 Quadratic Residues

The study of quadratic residues asks which congruence classes can occur as squares. For example, the squares modulo 7 are

$$0^2 \equiv 0, \quad 1^2 \equiv 6^2 \equiv 1, \quad 2^2 \equiv 5^2 \equiv 4, \quad 3^2 \equiv 4^2 \equiv 2 \pmod{7}.$$

Thus the nonzero squares modulo 7 are 1, 2, and 4, whereas 3, 5, and 6 are not squares. Throughout this section, p denotes an odd prime.

§12.1 Quadratic Residues and the Legendre Symbol

Let a be an integer with $p \nmid a$. We call a a *quadratic residue modulo p* if the congruence

$$x^2 \equiv a \pmod{p}$$

has an integer solution. Otherwise, a is called a *quadratic nonresidue modulo p* . These properties depend only on the residue class of a modulo p .

The *Legendre symbol* is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

From the list of squares modulo 7,

$$\left(\frac{1}{7}\right) = \left(\frac{2}{7}\right) = \left(\frac{4}{7}\right) = 1, \quad \left(\frac{3}{7}\right) = \left(\frac{5}{7}\right) = \left(\frac{6}{7}\right) = -1.$$

Also, $\left(\frac{7}{7}\right) = 0$.

There are exactly $(p-1)/2$ nonzero quadratic residues and $(p-1)/2$ quadratic nonresidues modulo p . Moreover, if $p \nmid a$ is a quadratic residue, then $x^2 \equiv a \pmod{p}$ has exactly two incongruent solutions modulo p .

Proof: Consider the squares of $1, 2, \dots, (p-1)/2$. If

$$x^2 \equiv y^2 \pmod{p},$$

then $p \mid (x-y)(x+y)$, so $x \equiv y$ or $x \equiv -y \pmod{p}$. For $1 \leq x, y \leq (p-1)/2$, the second possibility is impossible unless $x = y$, because $0 < x + y < p$. Hence these $(p-1)/2$ squares are pairwise incongruent.

Every nonzero square is represented in this list: if z is nonzero modulo p , then exactly one of z and $-z$ has a representative in $\{1, \dots, (p-1)/2\}$, and both have the same square. Thus there are exactly $(p-1)/2$ nonzero quadratic residues, leaving the same number of nonresidues.

Finally, if $x_0^2 \equiv a \pmod{p}$, then both x_0 and $-x_0$ are solutions. Any other solution x satisfies $p \mid (x-x_0)(x+x_0)$, so $x \equiv x_0$ or $x \equiv -x_0 \pmod{p}$. Since $p \nmid x_0$ and p is odd, these two solutions are distinct. $\square$

§12.2 Euler's Criterion

Euler's criterion:

For every integer a and every odd prime p ,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

In particular, when $p \nmid a$,

$$a^{(p-1)/2} \equiv \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

Proof: If $p \mid a$, both sides are congruent to 0 modulo p . Now suppose $p \nmid a$ and put $h = (p-1)/2$.

If a is a quadratic residue, write $a \equiv x^2 \pmod{p}$ with $p \nmid x$. Fermat's little theorem gives

$$a^h \equiv x^{p-1} \equiv 1 \pmod{p}.$$

The polynomial $T^h - 1$ has degree h , so it has at most h roots modulo p . The preceding argument shows that all h nonzero quadratic residues are roots; therefore they are exactly its roots. If a is a quadratic nonresidue, then $a^h \not\equiv 1 \pmod{p}$. On the other hand, Fermat's little theorem gives

$$(a^h)^2 = a^{p-1} \equiv 1 \pmod{p}.$$

Thus $p \mid (a^h - 1)(a^h + 1)$. Since the first factor is not divisible by p , the second one is, and hence $a^h \equiv -1 \pmod{p}$. This proves the criterion. $\square$

The Legendre symbol is multiplicative in its numerator:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$$

Proof: Euler's criterion gives

$$\left(\frac{ab}{p}\right) \equiv (ab)^{(p-1)/2} = a^{(p-1)/2} b^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) \pmod{p}.$$

Both sides belong to $\{-1, 0, 1\}$, so the congruence is an equality. $\square$

The first supplementary law is

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Thus -1 is a quadratic residue modulo p exactly when $p \equiv 1 \pmod{4}$.

Proof: Apply Euler's criterion with $a = -1$. $\square$

§12.3 Gauss's Lemma

Gauss's lemma: Let $p \nmid a$ and put $h = (p-1)/2$. For each of

$$a, 2a, \dots, ha,$$

choose its unique representative in the interval $(-p/2, p/2)$. If ν of these representatives are negative, then

$$\left(\frac{a}{p}\right) = (-1)^\nu.$$

Proof: Let the chosen signed representatives be $r_1, \dots, r_h$. Their absolute values are a permutation of $1, 2, \dots, h$. Indeed, if $|r_i| = |r_j|$, then

$$ai \equiv \pm aj \pmod{p}.$$

Since $p \nmid a$, cancellation gives $i \equiv \pm j \pmod{p}$. The plus sign gives $i = j$, while the minus sign would imply $p \mid i + j$, which is impossible because $2 \leq i + j \leq p-1$.

Multiplying the congruences $ak \equiv r_k \pmod{p}$ gives

$$a^h h! \equiv r_1 \cdots r_h = (-1)^\nu h! \pmod{p}.$$

Since none of $1, \dots, h$ is divisible by p , we may cancel $h!$:

$$a^{(p-1)/2} \equiv (-1)^\nu \pmod{p}.$$

Euler's criterion now yields $\left(\frac{a}{p}\right) = (-1)^\nu$. $\square$

The second supplementary law:

For every odd prime p ,

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8}, \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8}. \end{cases}$$

Proof: Apply Gauss's lemma with $a = 2$ and $h = (p-1)/2$. The numbers $2, 4, \dots, 2h$ all lie between 1 and $p-1$. The representative of $2k$ in $(-p/2, p/2)$ is negative exactly when $2k > p/2$, or $k > p/4$. Hence

$$\nu = h - \left\lfloor \frac{p}{4} \right\rfloor.$$

Checking the four possible odd residue classes of p modulo 8 shows that ν is even for $p \equiv 1, 7 \pmod{8}$ and odd for $p \equiv 3, 5 \pmod{8}$. Equivalently,

$$\nu \equiv \frac{p^2-1}{8} \pmod{2}.$$

Gauss's lemma therefore gives the stated formula. $\square$

§12.4 The Law of Quadratic Reciprocity

Eisenstein's form of Gauss's lemma: Let p be an odd prime and let a be a positive odd integer with $p \nmid a$. Then

$$\left(\frac{a}{p}\right) = (-1)^{\sum_{j=1}^{(p-1)/2} \left\lfloor \frac{aj}{p} \right\rfloor}.$$

Proof: Put $h = (p-1)/2$. For $1 \leq j \leq h$, write

$$aj = pq_j + r_j, \quad q_j = \left\lfloor \frac{aj}{p} \right\rfloor, \quad 1 \leq r_j \leq p-1.$$

Let $s_j = r_j$ when $r_j < p/2$, and let $s_j = p - r_j$ when $r_j > p/2$. As in the proof of Gauss's lemma, the numbers $s_1, \dots, s_h$ are a permutation of $1, \dots, h$. Let ν be the number of indices for which $r_j > p/2$. Then

$$\sum_{j=1}^h r_j = \sum_{j=1}^h s_j + \nu p - 2 \sum_{r_j > p/2} s_j.$$

Reducing modulo 2 and using that p and a are odd gives

$$\begin{aligned} \sum_{j=1}^h j &\equiv a \sum_{j=1}^h j = p \sum_{j=1}^h q_j + \sum_{j=1}^h r_j \\ &\equiv \sum_{j=1}^h q_j + \sum_{j=1}^h s_j + \nu \equiv \sum_{j=1}^h q_j + \sum_{j=1}^h j + \nu \pmod{2}. \end{aligned}$$

Therefore

$$\nu \equiv \sum_{j=1}^h q_j = \sum_{j=1}^h \left\lfloor \frac{aj}{p} \right\rfloor \pmod{2}.$$

Gauss's lemma now gives the stated formula. $\square$

Law of quadratic reciprocity:

Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless $p \equiv q \equiv 3 \pmod{4}$, in which case

$$\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right).$$

Proof: Put

$$m = \frac{p-1}{2}, \quad n = \frac{q-1}{2}.$$

Eisenstein's form of Gauss's lemma gives

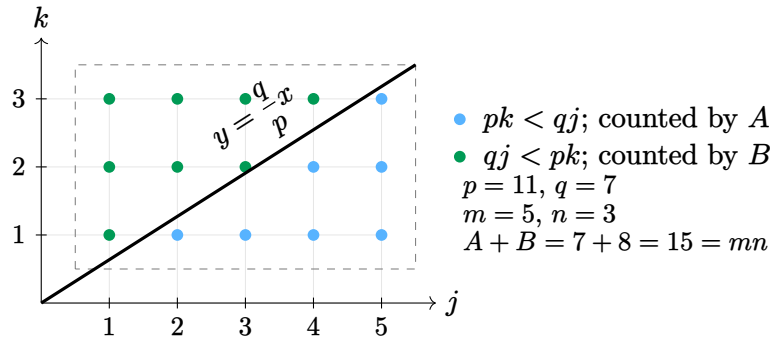
$$\left(\frac{q}{p}\right) = (-1)^A, \quad A = \sum_{j=1}^m \left\lfloor \frac{jq}{p} \right\rfloor,$$

and

$$\left(\frac{p}{q}\right) = (-1)^B, \quad B = \sum_{k=1}^n \left\lfloor \frac{kp}{q} \right\rfloor.$$

Consider the mn lattice points (j, k) satisfying

$$1 \leq j \leq m, \quad 1 \leq k \leq n.$$



The case $p = 11$ and $q = 7$: each of the $mn = 15$ lattice points lies strictly below or strictly above the line.

The sum A counts the points for which $pk < qj$, while B counts the points for which $qj < pk$. Equality cannot occur: if $pk = qj$, then the distinct primes p and q would force $p \mid j$, which is impossible because $1 \leq j < p$. Hence every point is counted exactly once by either A or B , and therefore

$$A + B = mn = \frac{p-1}{2} \cdot \frac{q-1}{2}$$

Consequently,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{A+B} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

which proves the law. $\square$

For example, quadratic reciprocity gives

$$\left(\frac{3}{11}\right) = -\left(\frac{11}{3}\right) = -\left(\frac{2}{3}\right) = 1.$$

Indeed, $5^2 = 25 \equiv 3 \pmod{11}$.